

Functions and Function Notation (Part IV) Homework

The graphs of various functions are written below. Write the domain and range of each function in words, set notation, and interval notation.

① Domain

Words:

Set Notation:

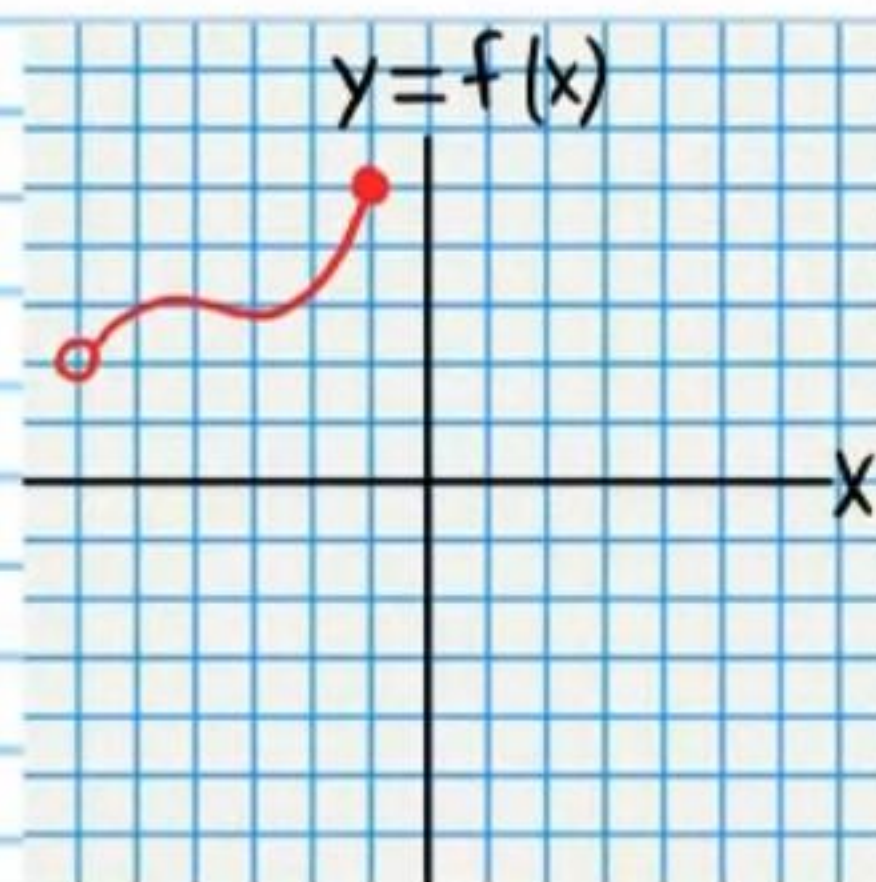
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



② Domain

Words:

Set Notation:

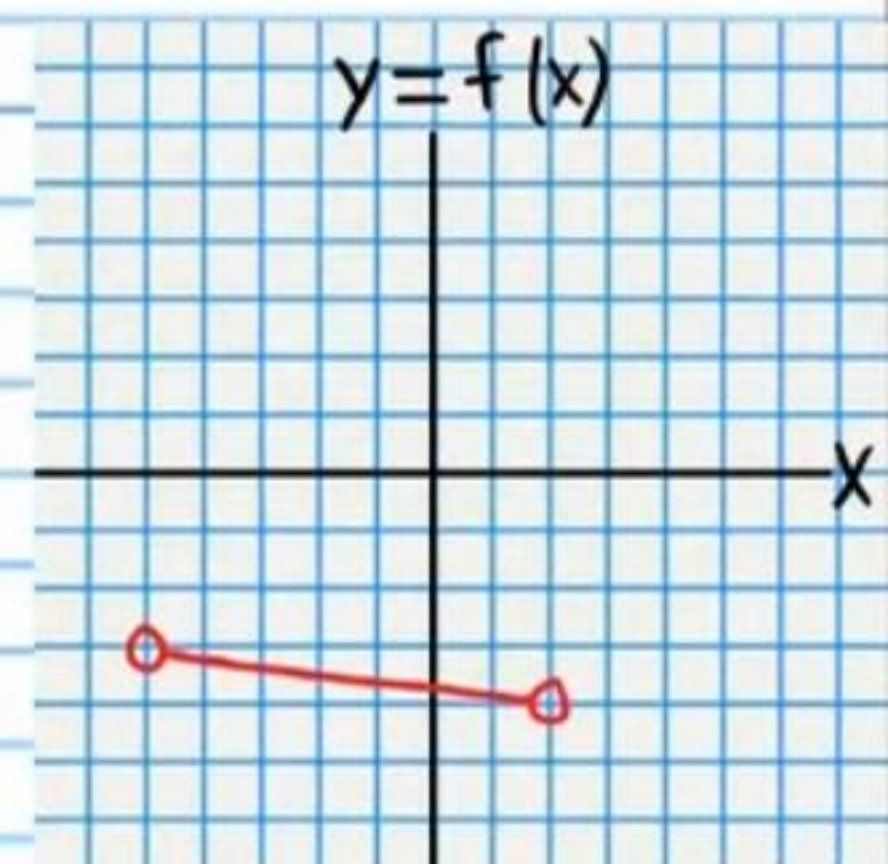
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



③ Domain

Words:

Set Notation:

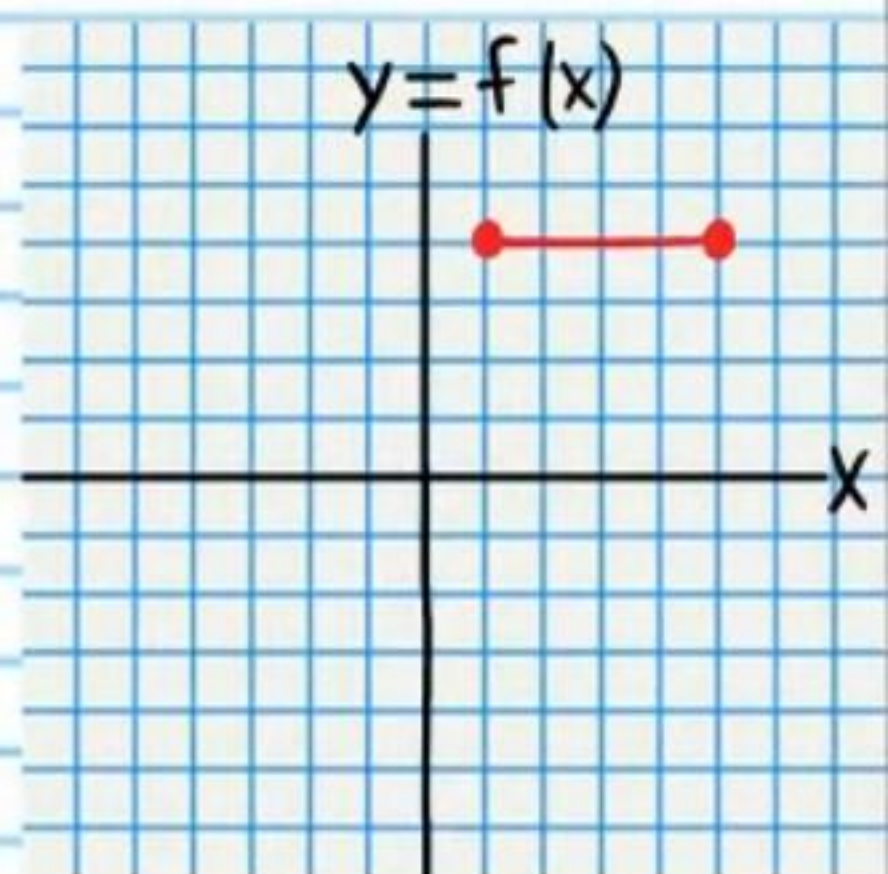
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



④ Domain

Words:

Set Notation:

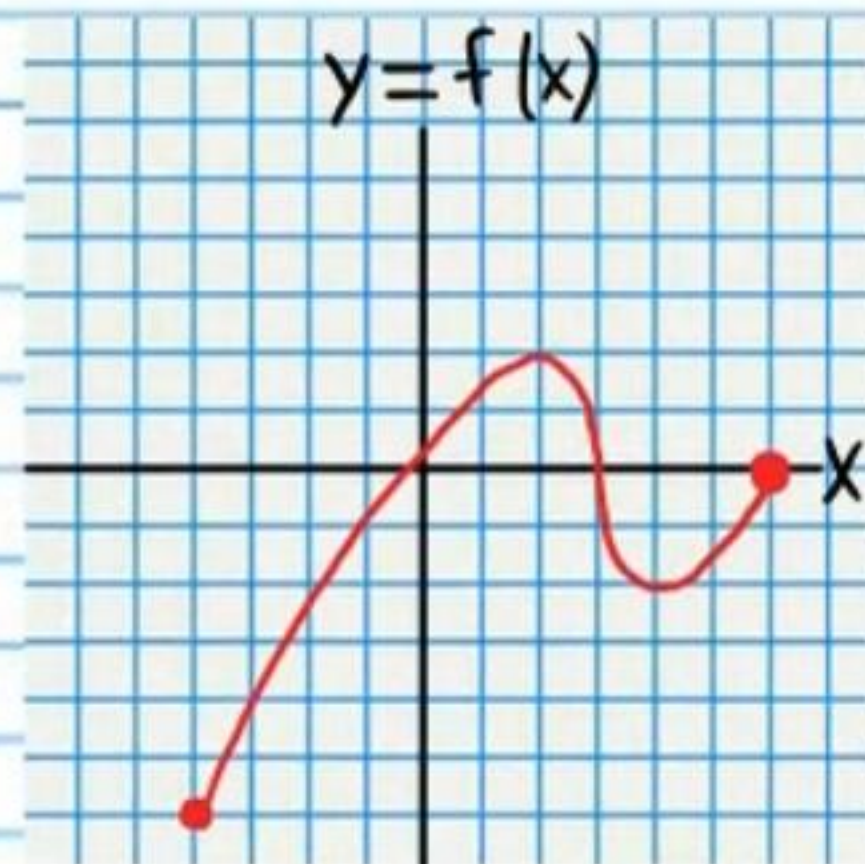
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑤ Domain

Words:

Set Notation:

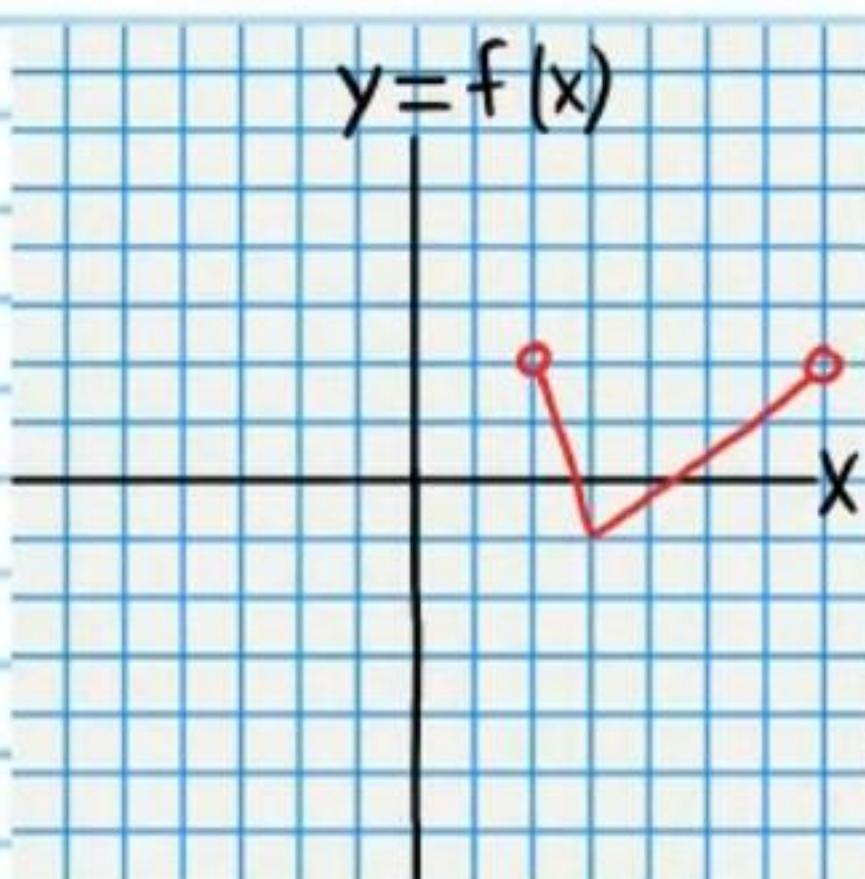
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑥ Domain

Words:

Set Notation:

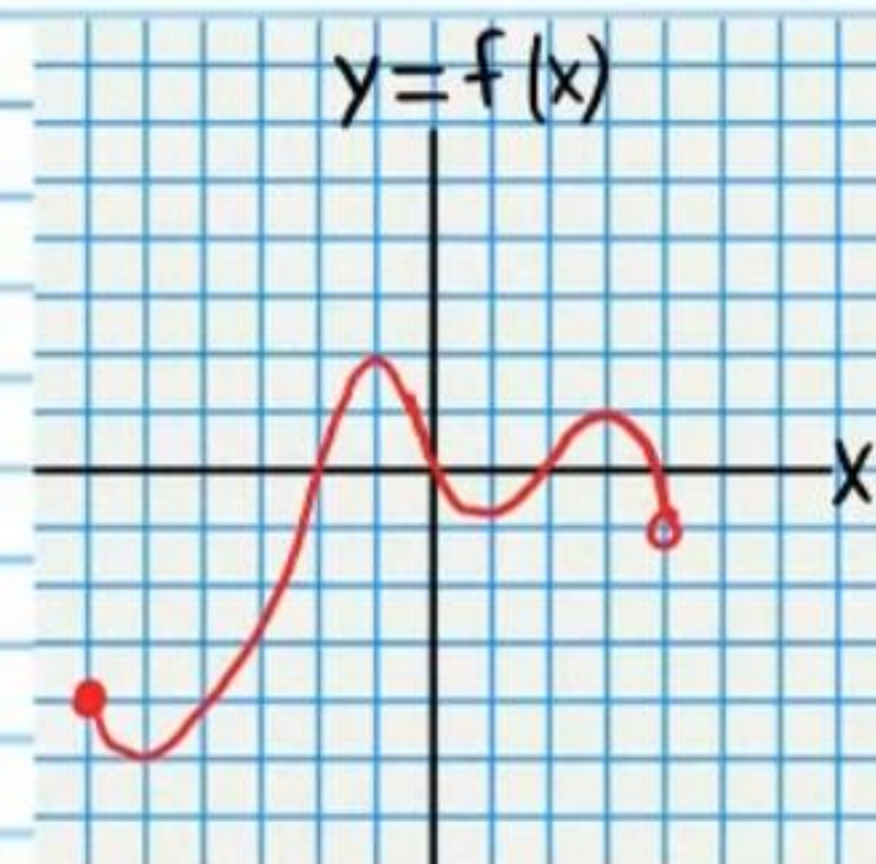
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑦ Domain

Words:

Set Notation:

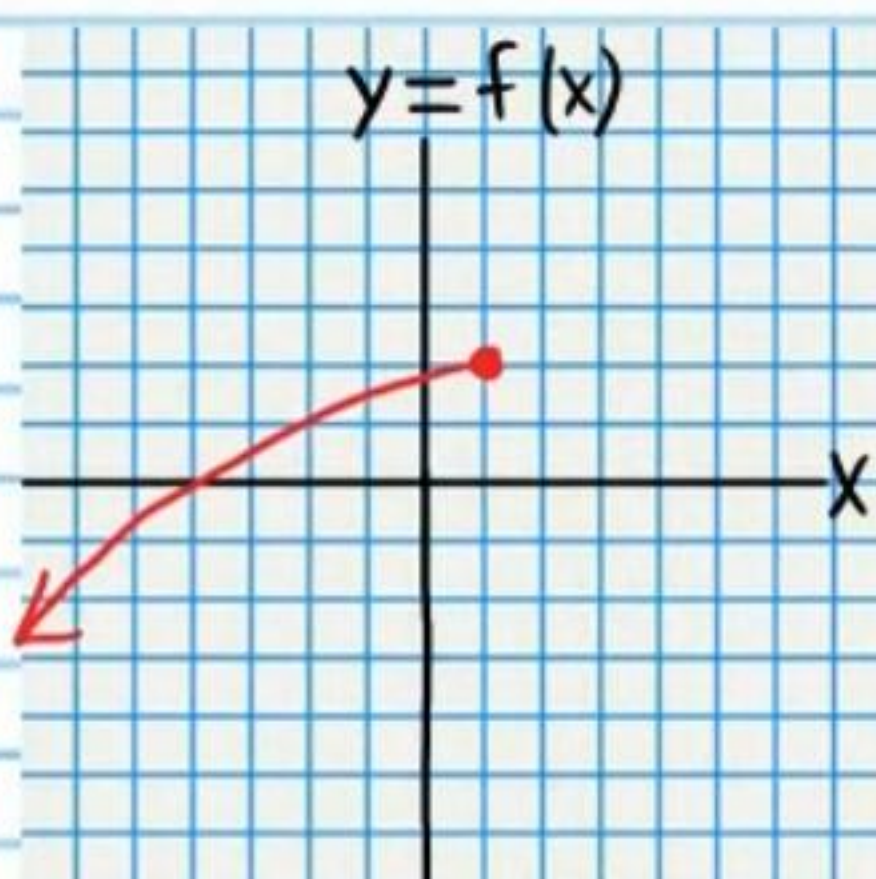
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑧ Domain

Words:

Set Notation:

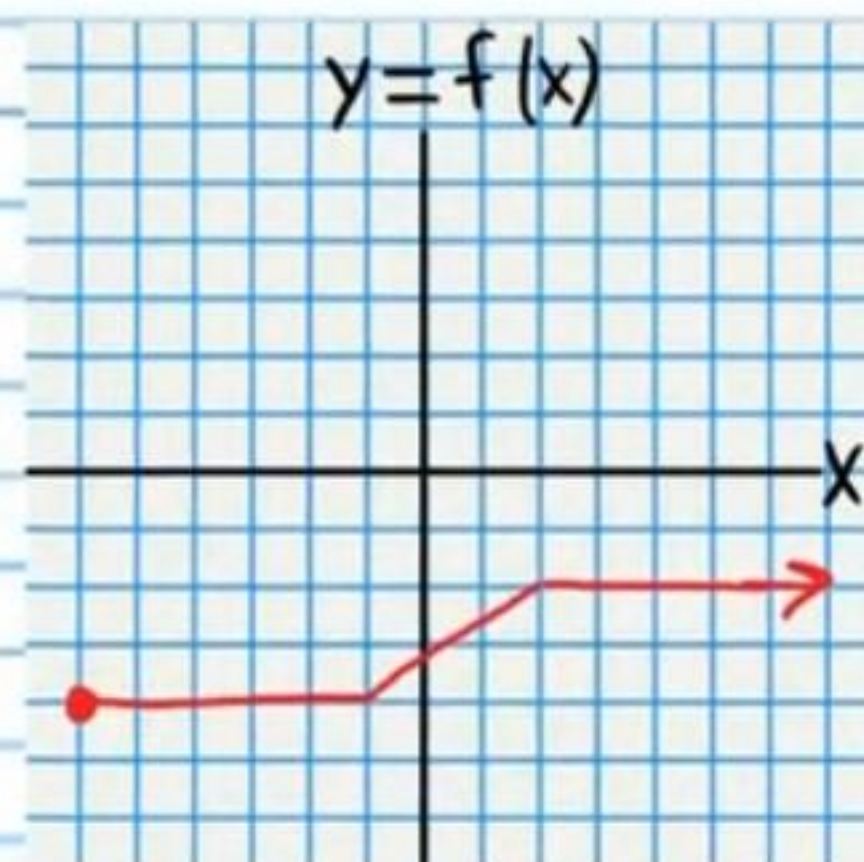
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑨ Domain

Words:

Set Notation:

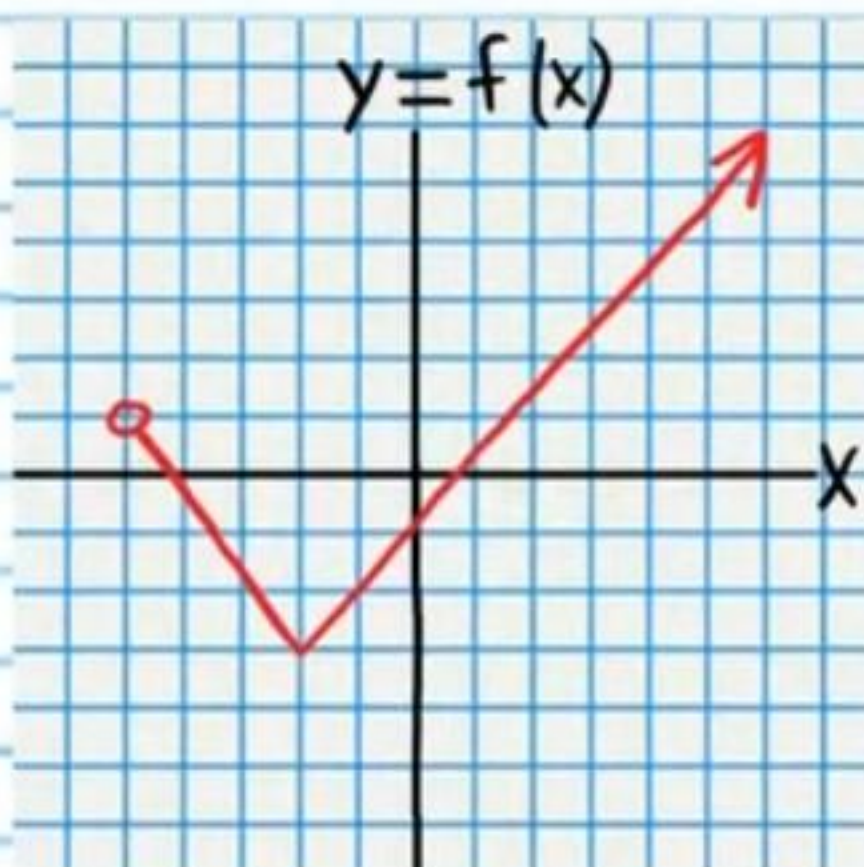
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑩ Domain

Words:

Set Notation:

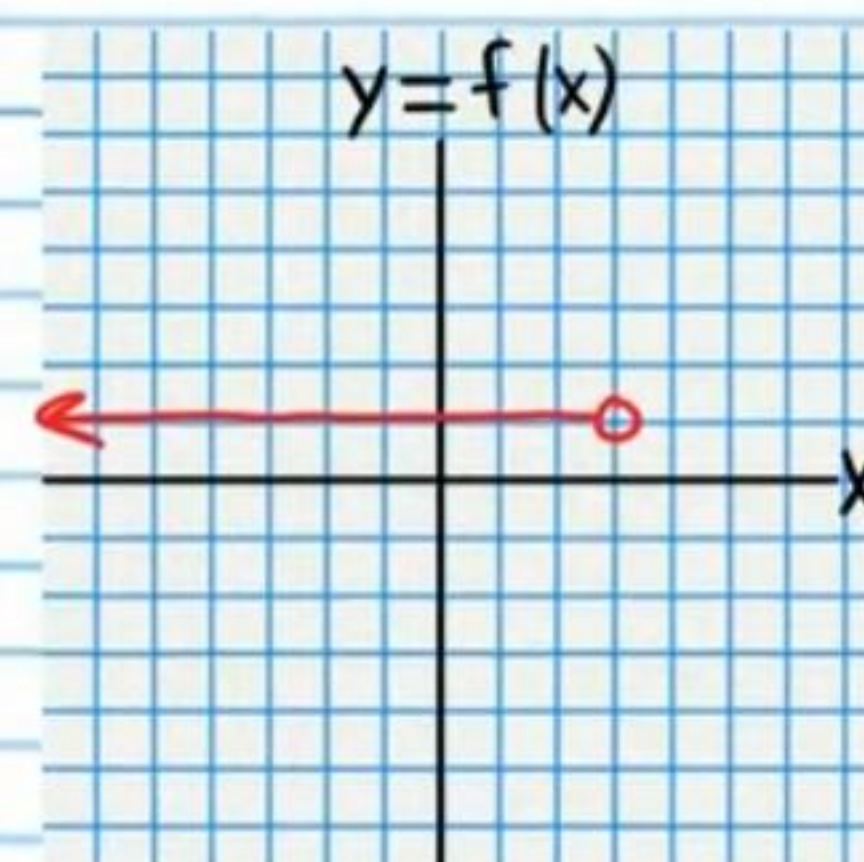
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑪ Domain

Words:

Set Notation:

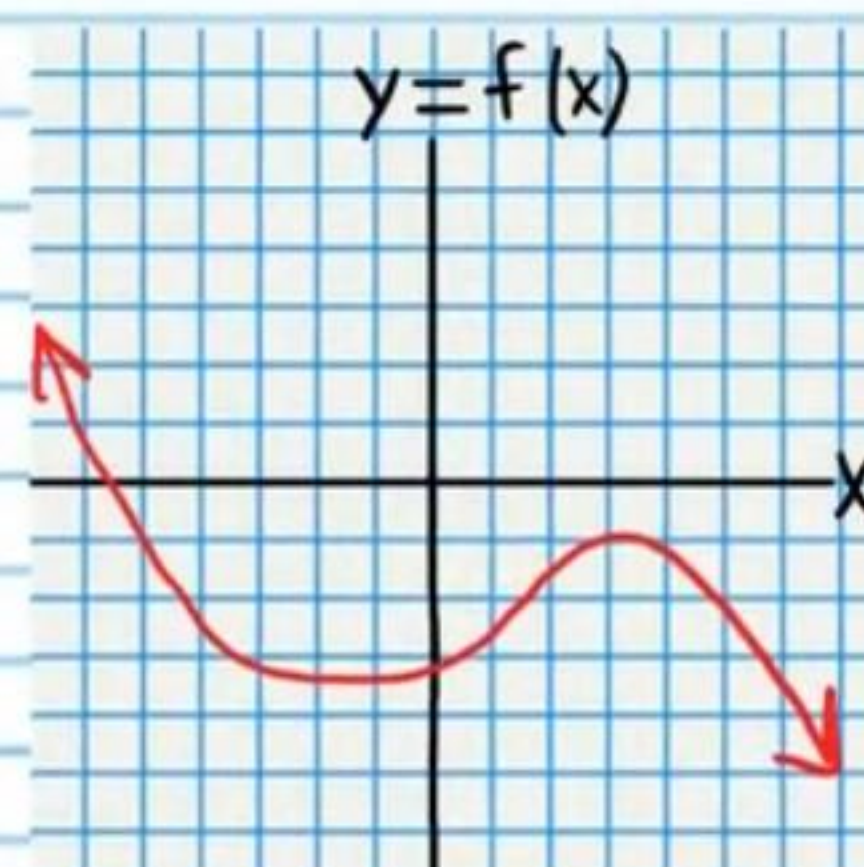
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑫ Domain

Words:

Set Notation:

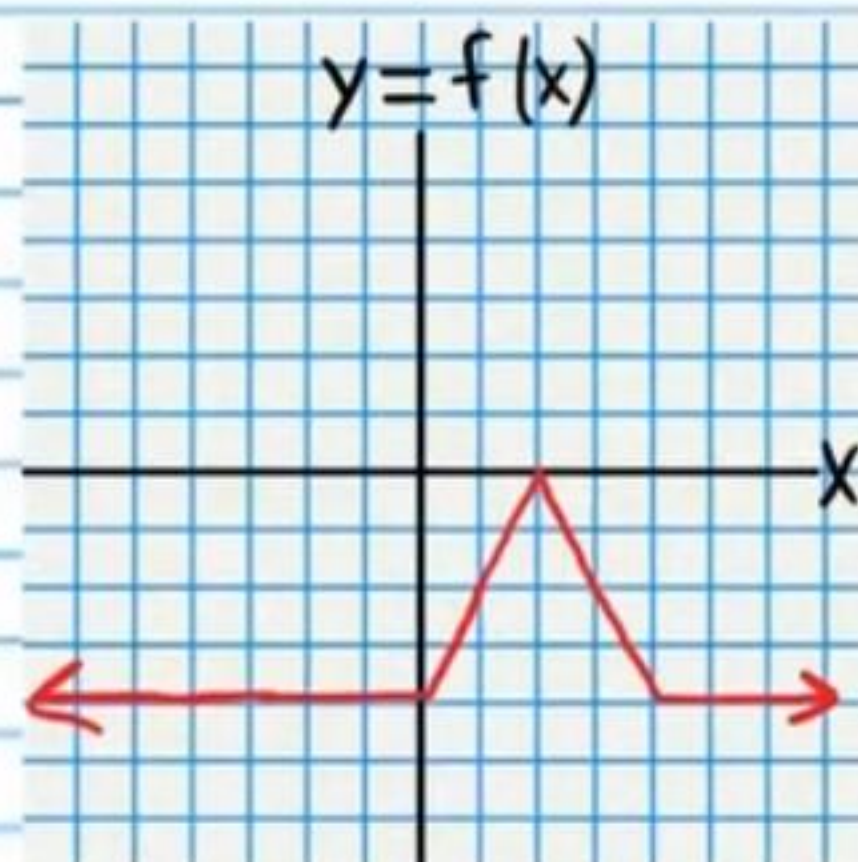
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑬ Domain

Words:

Set Notation:

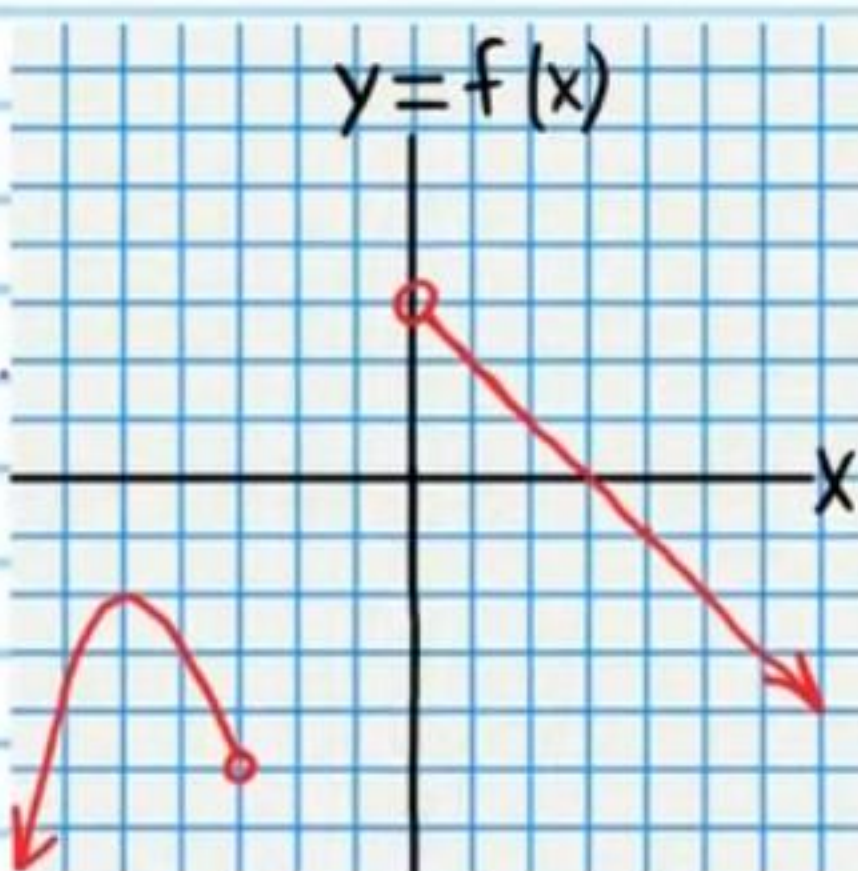
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑭ Domain

Words:

Set Notation:

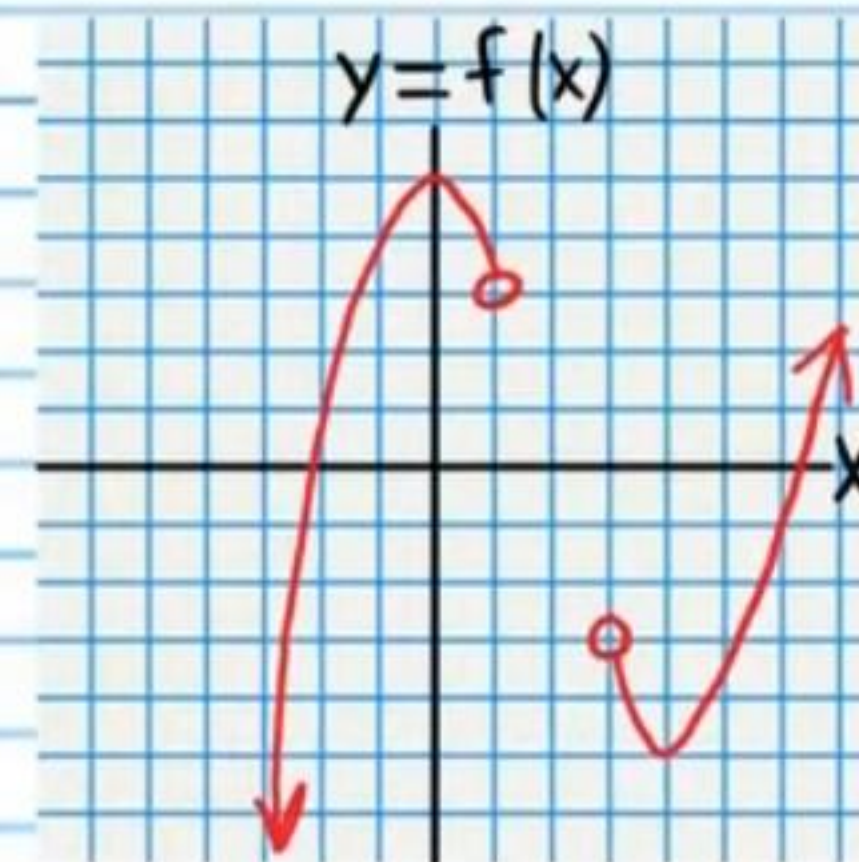
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑮ Domain

Words:

Set Notation:

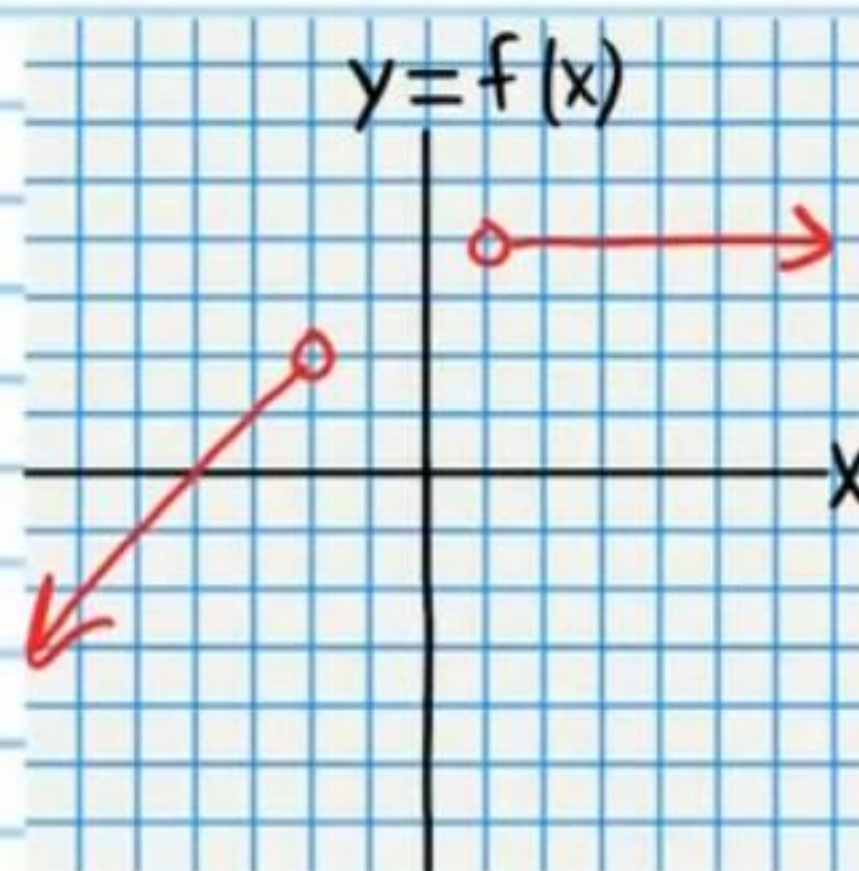
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑩ Domain

Words:

Set Notation:

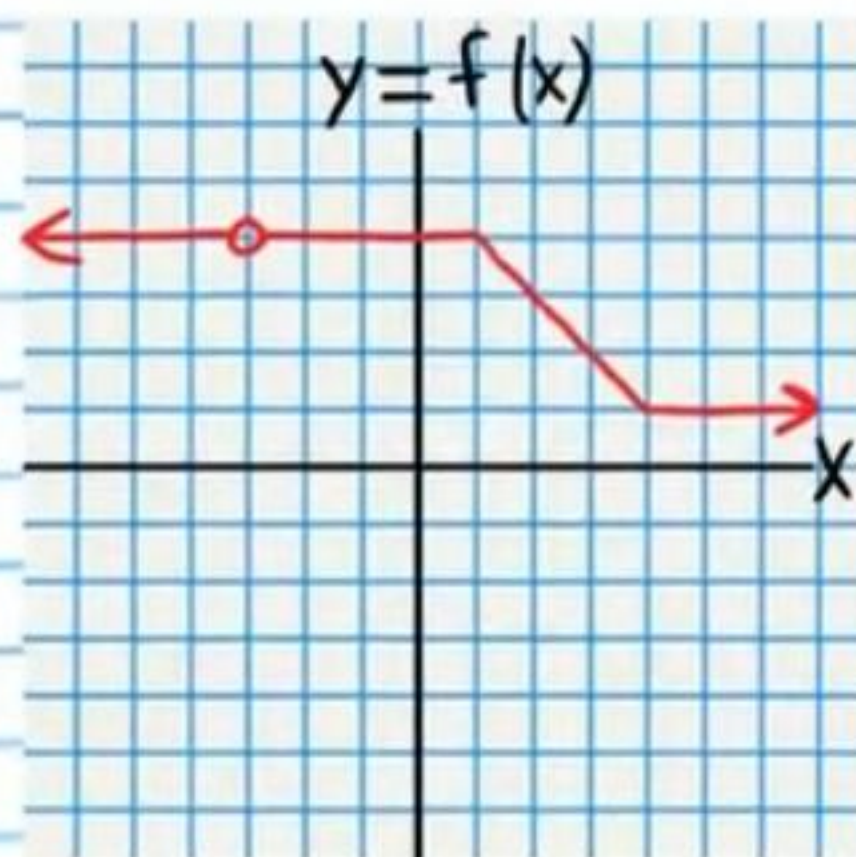
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑪ Domain

Words:

Set Notation:

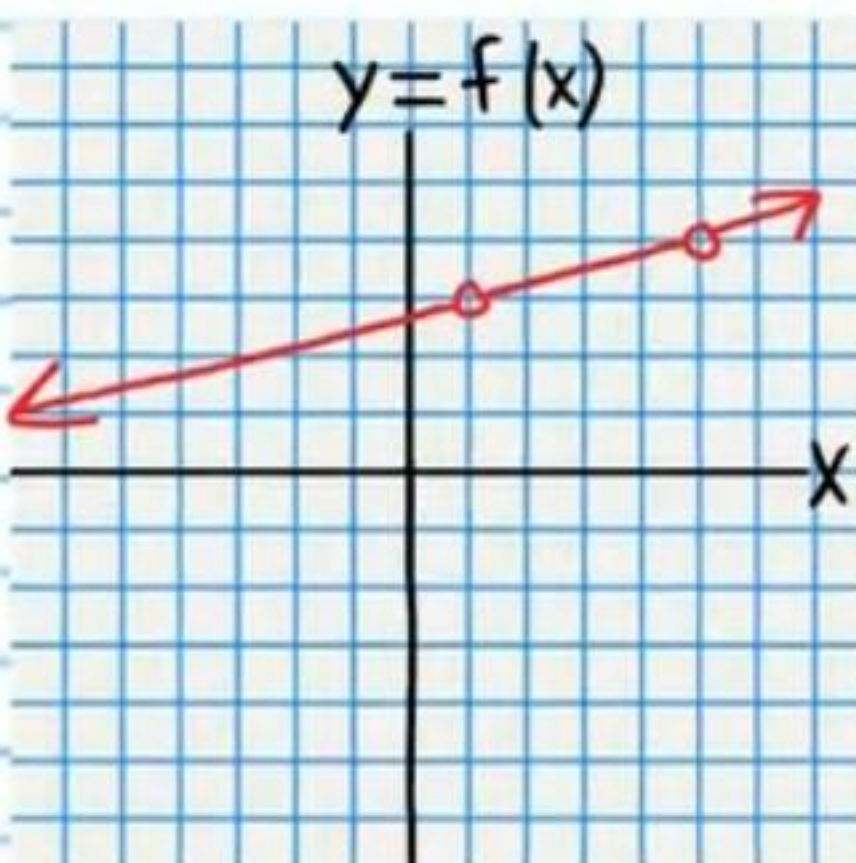
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑫ Domain

Words:

Set Notation:

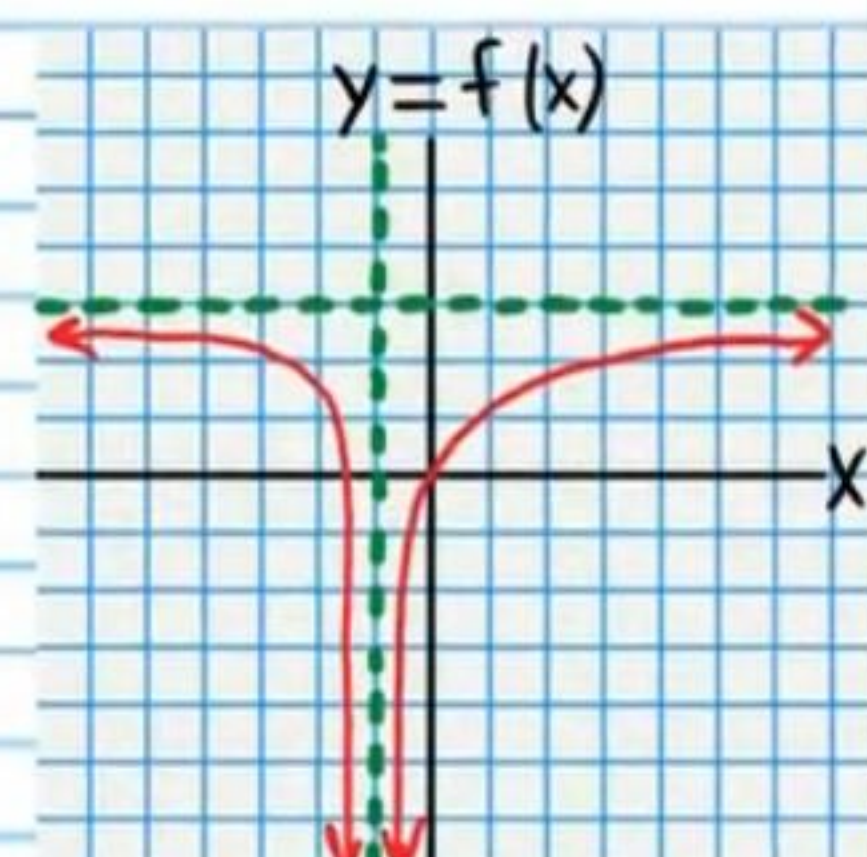
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑬ Domain

Words:

Set Notation:

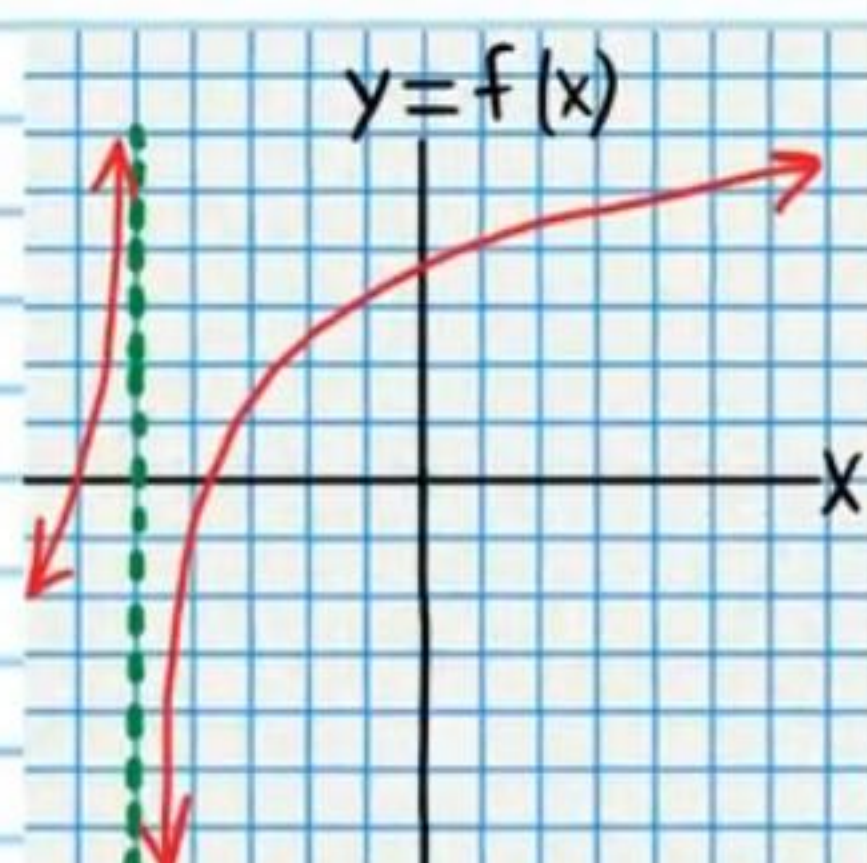
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



②① Domain

Words:

Set Notation:

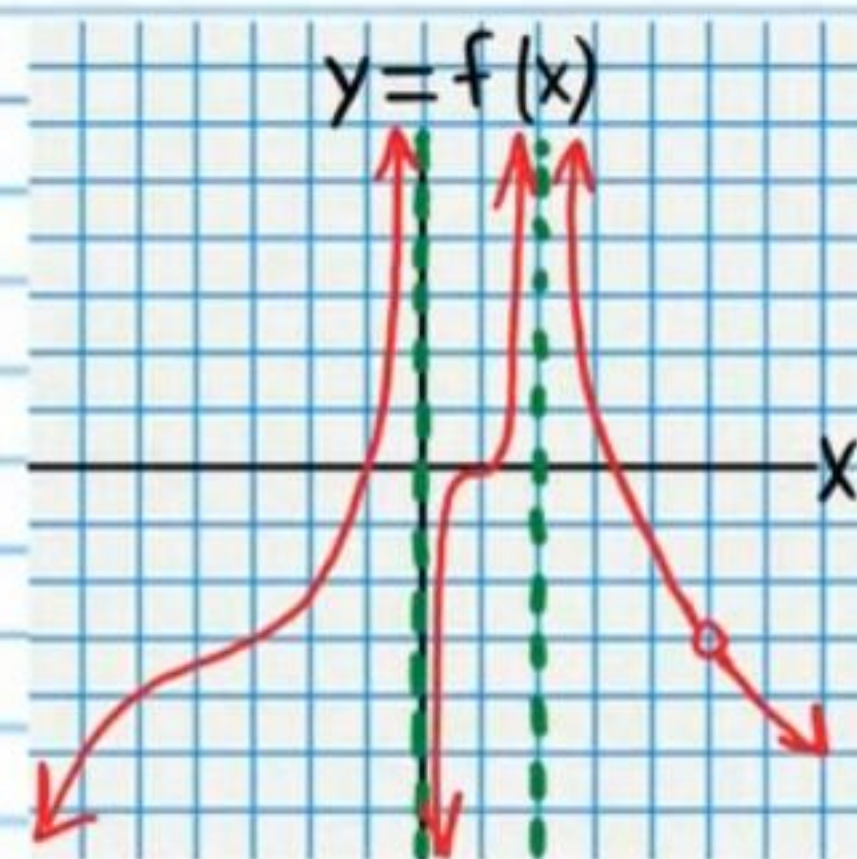
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



②② Domain

Words:

Set Notation:

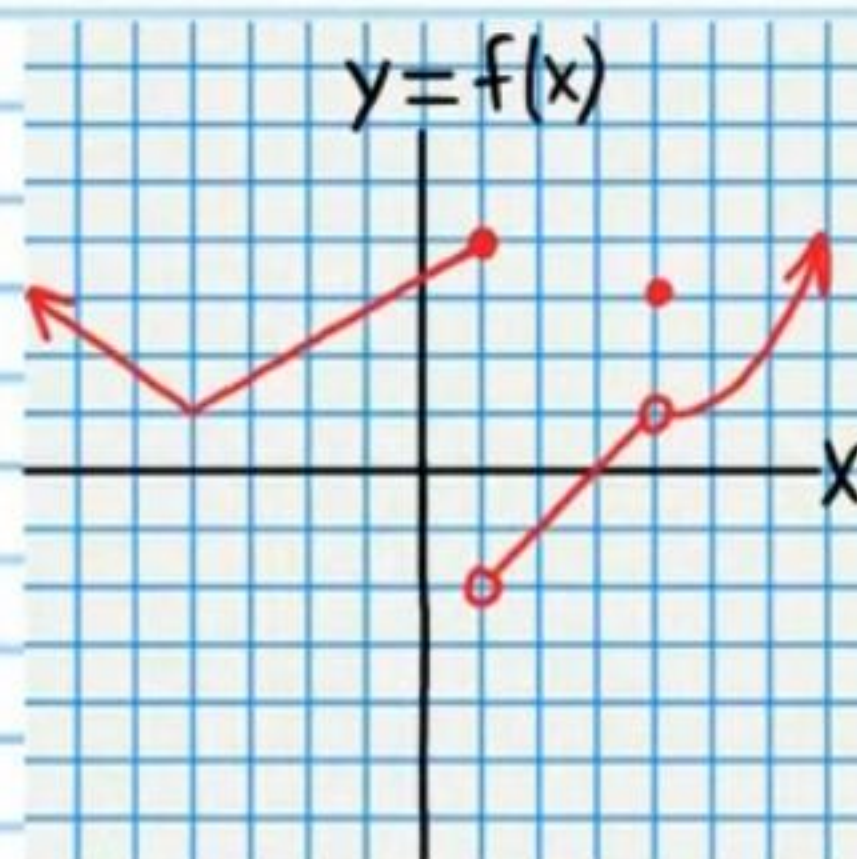
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



②③ Domain

Words:

Set Notation:

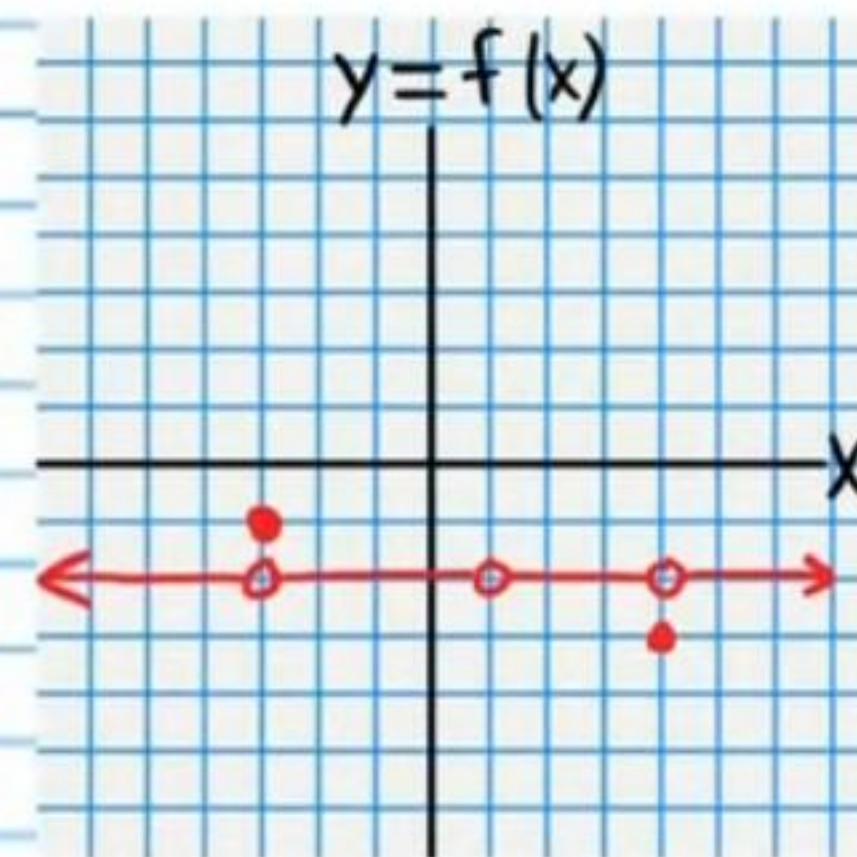
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



②④ Domain

Words:

Set Notation:

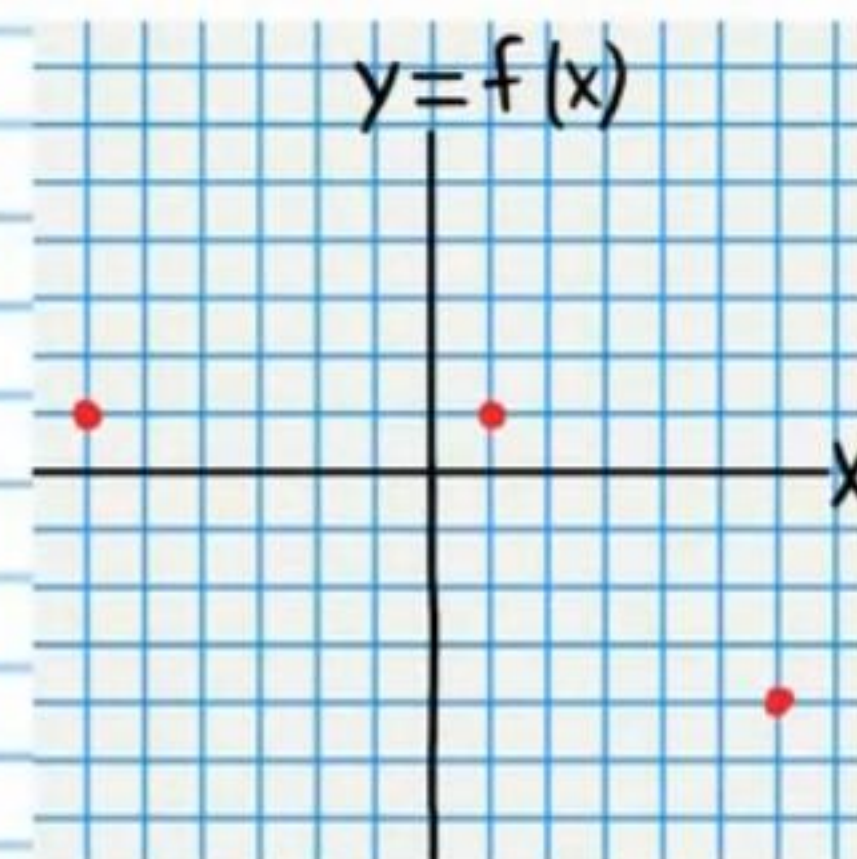
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



Answers

① Domain

Words: All real numbers between -6 and -1, including -1.

Set Notation: $\{x \mid -6 < x \leq -1\}$

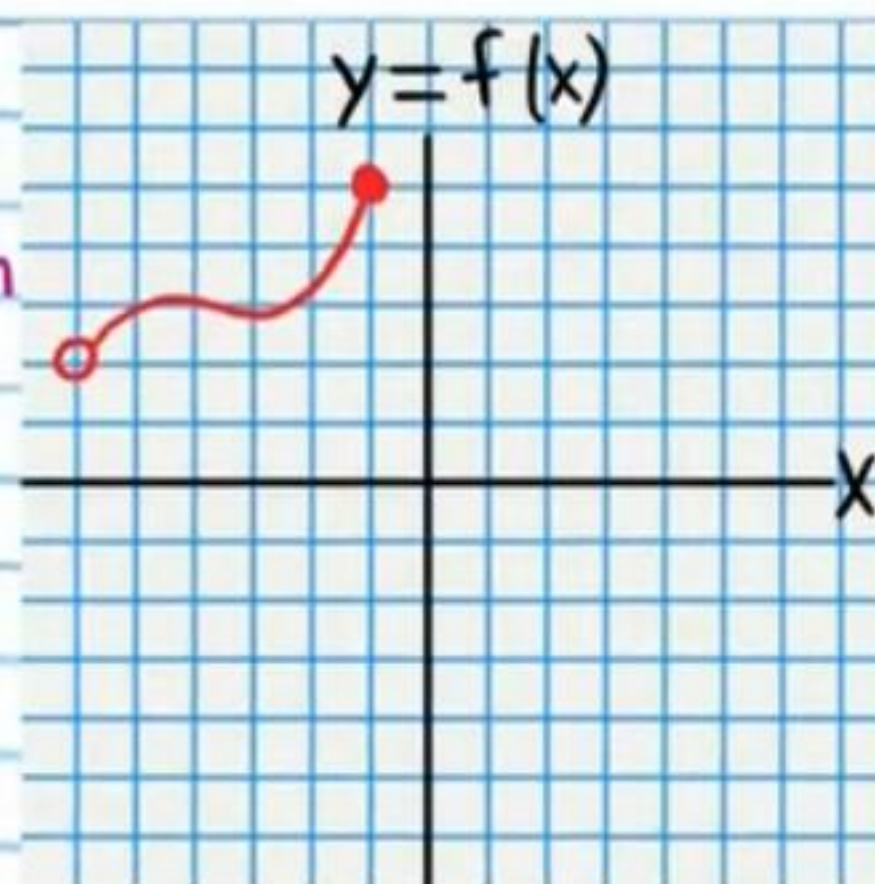
Interval Notation: $(-6, -1]$

Range

Words: All real numbers between 2 and 5, including 5.

Set Notation: $\{y \mid 2 < y \leq 5\}$

Interval Notation: $(2, 5]$



② Domain

Words: All real numbers between -5 and 2.

Set Notation: $\{x \mid -5 < x < 2\}$

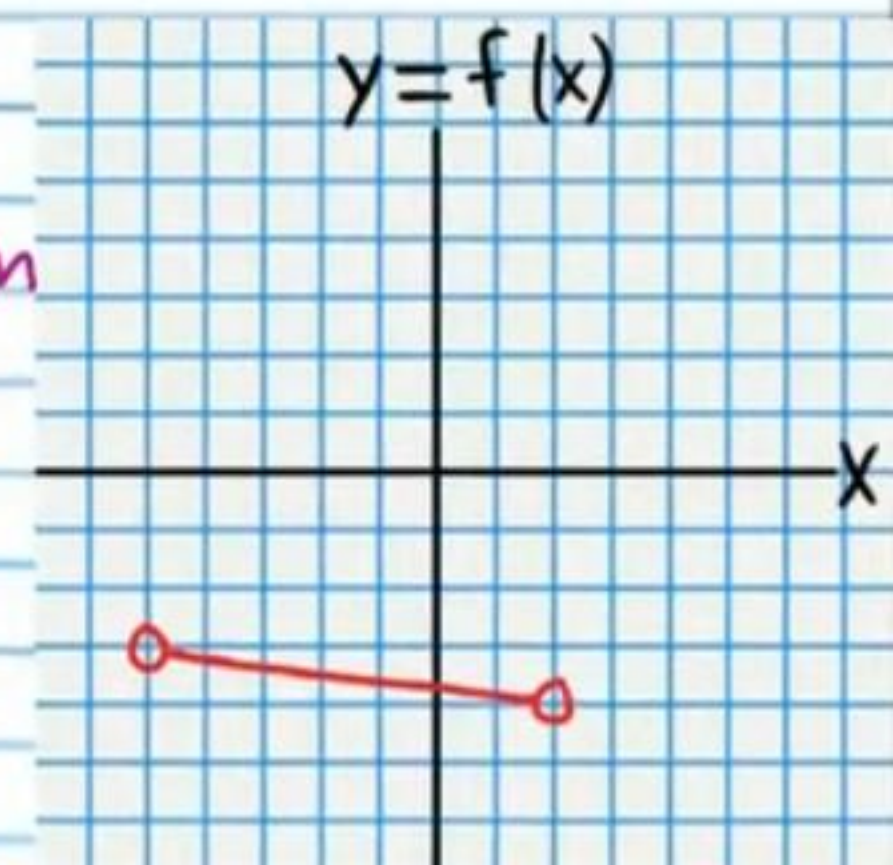
Interval Notation: $(-5, 2)$

Range

Words: All real numbers between -4 and -3.

Set Notation: $\{y \mid -4 < y < -3\}$

Interval Notation: $(-4, -3)$



③ Domain

Words: All real numbers between 1 and 5, inclusive.

Set Notation: $\{x \mid 1 \leq x \leq 5\}$

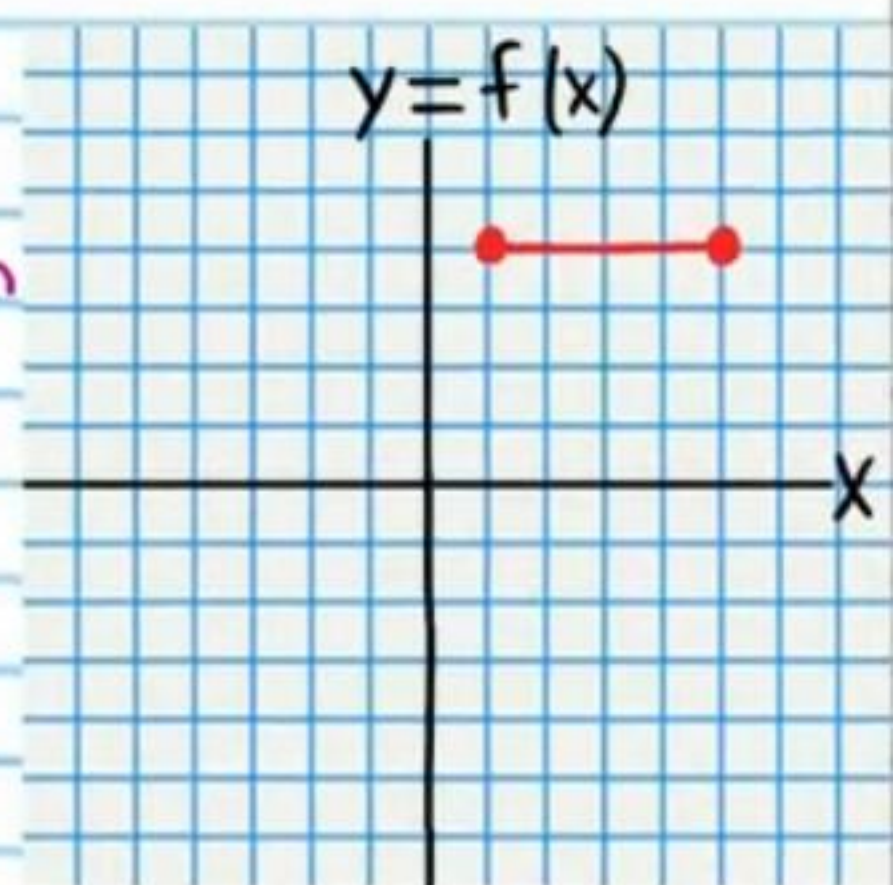
Interval Notation: $[1, 5]$

Range

Words: The only number in the range is 4.

Set Notation: $\{y \mid y = 4\}$

Interval Notation: $[4, 4]$

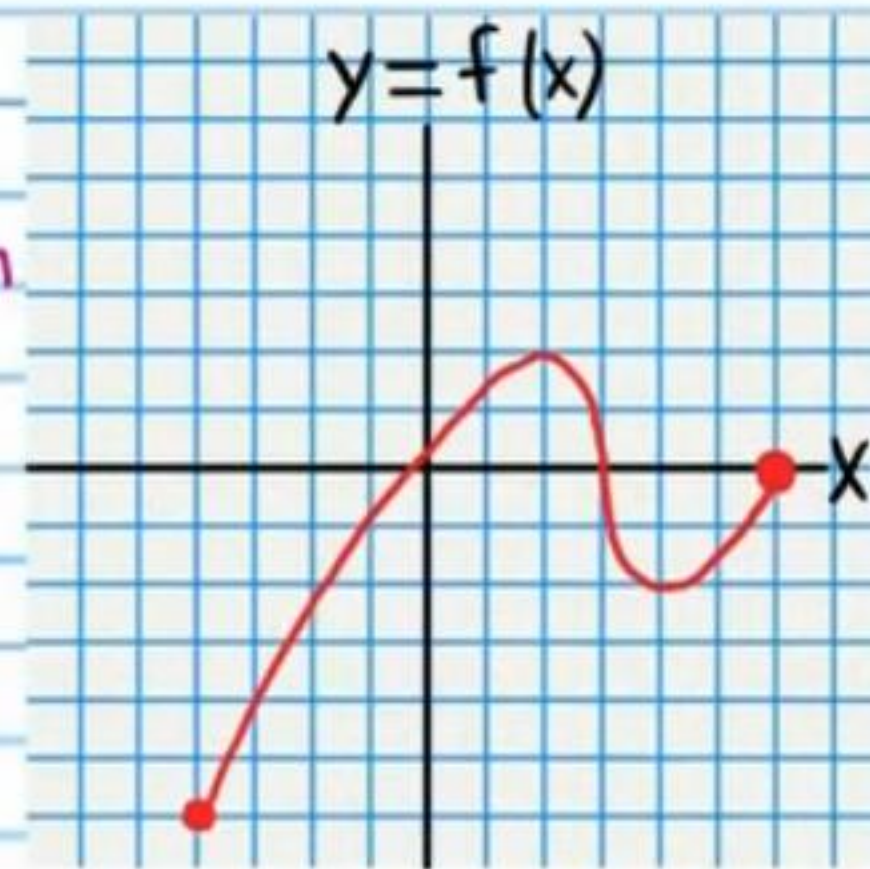


④ Domain

Words: All real numbers between -4 and 6, inclusive.

Set Notation: $\{x \mid -4 \leq x \leq 6\}$

Interval Notation: $[-4, 6]$



Range

Words: All real numbers between -6 and 2, inclusive.

Set Notation: $\{y \mid -6 \leq y \leq 2\}$

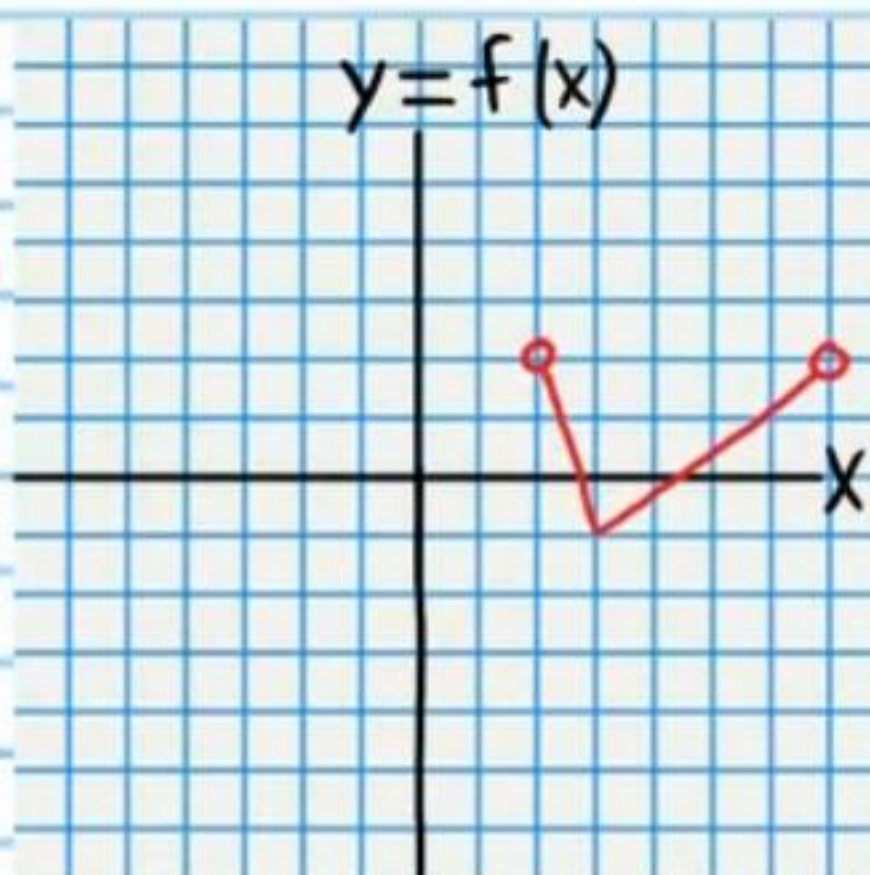
Interval Notation: $[-6, 2]$

⑤ Domain

Words: All real numbers between 2 and 7.

Set Notation: $\{x \mid 2 < x < 7\}$

Interval Notation: $(2, 7)$



Range

Words: All real numbers between -1 and 2, including -1.

Set Notation: $\{y \mid -1 \leq y < 2\}$

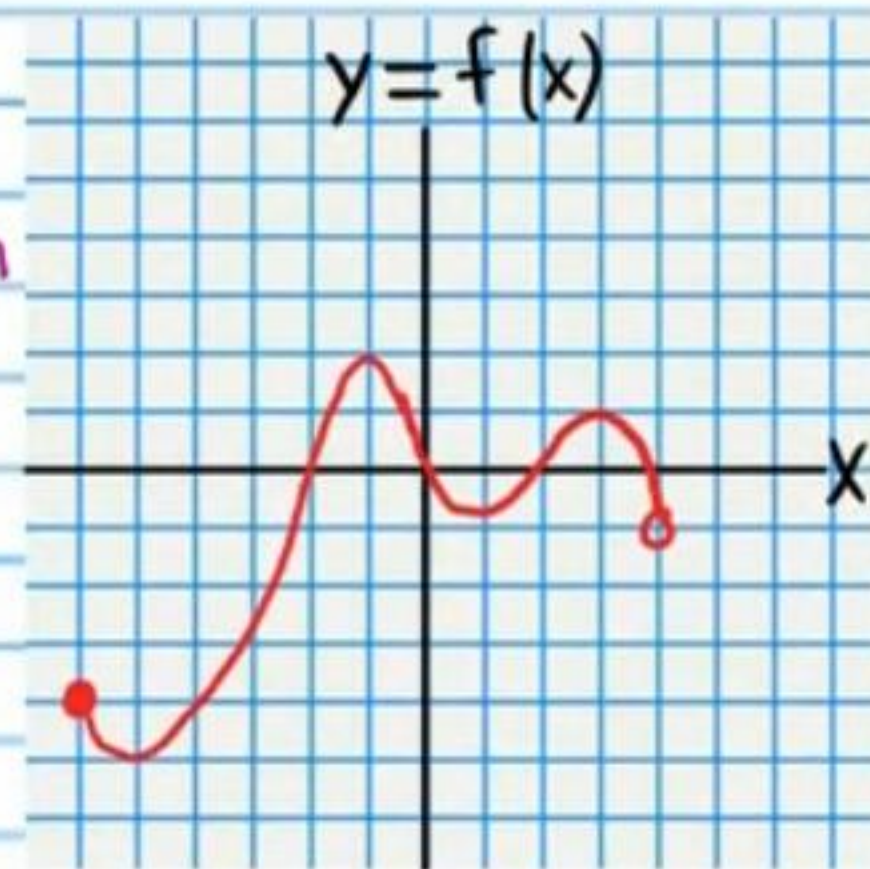
Interval Notation: $[-1, 2)$

⑥ Domain

Words: All real numbers between -6 and 4, including -6.

Set Notation: $\{x \mid -6 \leq x < 4\}$

Interval Notation: $[-6, 4)$



Range

Words: All real numbers between -5 and 2, inclusive.

Set Notation: $\{y \mid -5 \leq y \leq 2\}$

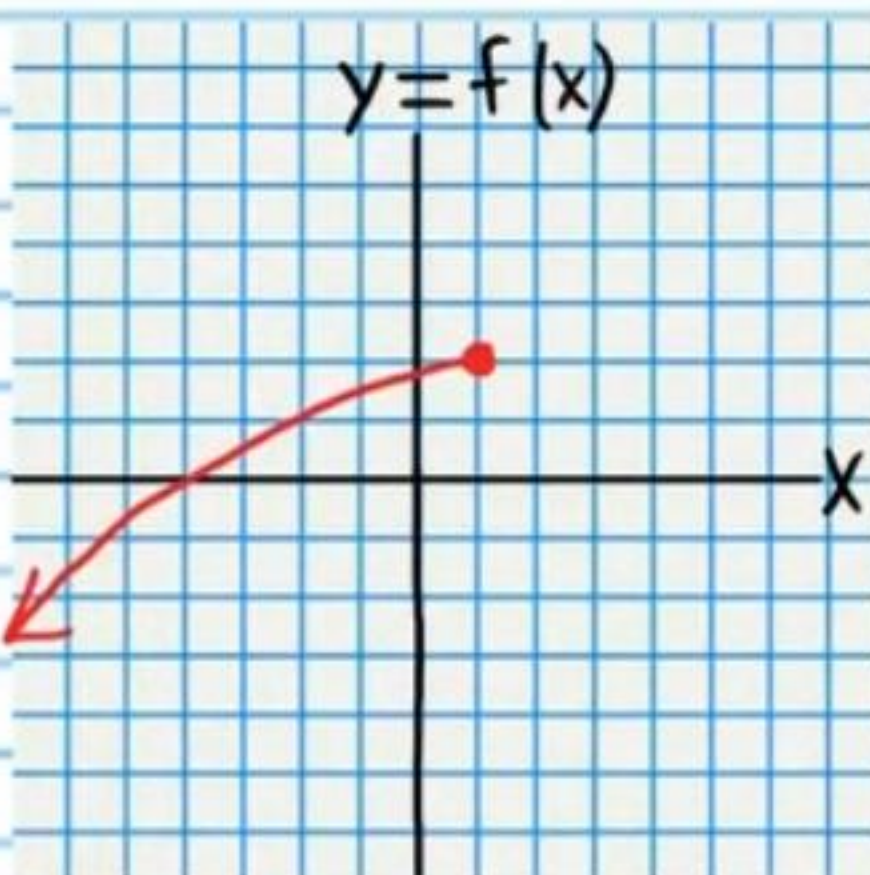
Interval Notation: $[-5, 2]$

⑦ Domain

Words: All real numbers less than or equal to 1.

Set Notation: $\{x \mid x \leq 1\}$

Interval Notation: $(-\infty, 1]$



Range

Words: All real numbers less than or equal to 2.

Set Notation: $\{y \mid y \leq 2\}$

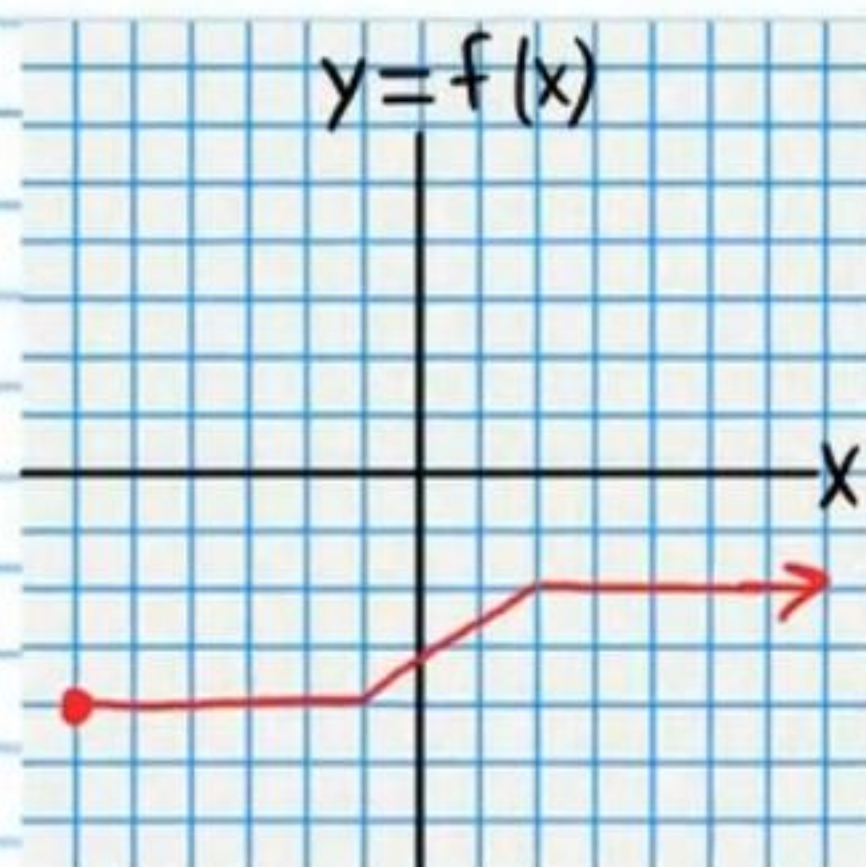
Interval Notation: $(-\infty, 2]$

⑧ Domain

Words: All real numbers greater than or equal to -6.

Set Notation: $\{x \mid x \geq -6\}$

Interval Notation: $[-6, \infty)$



Range

Words: All real numbers between -4 and -2, inclusive.

Set Notation: $\{y \mid -4 \leq y \leq -2\}$

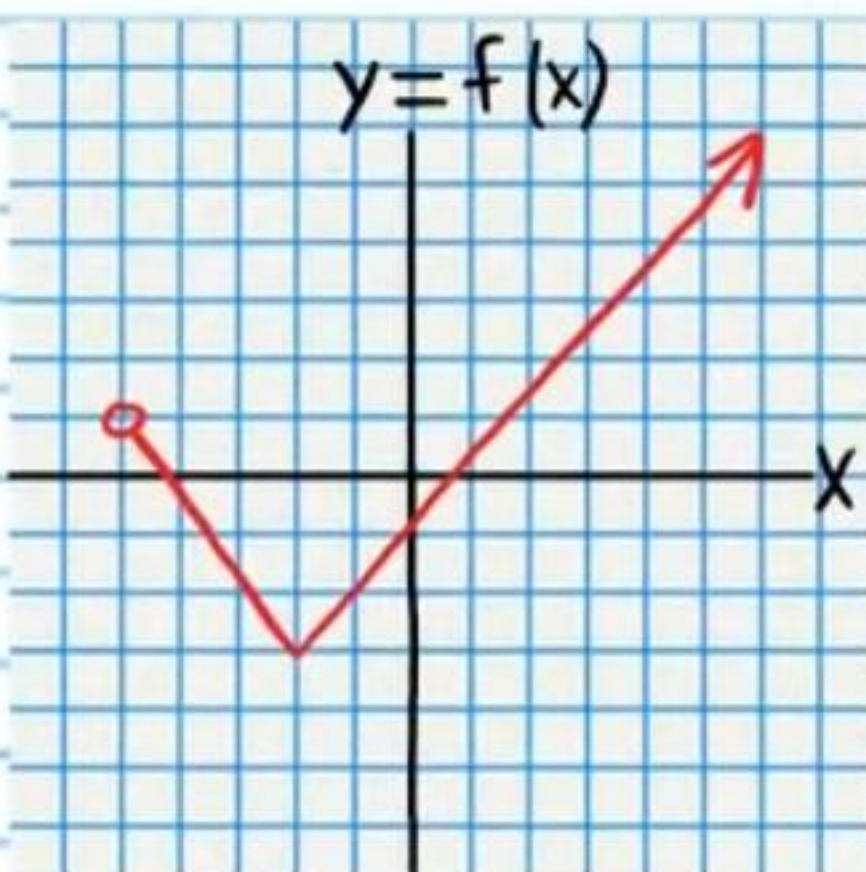
Interval Notation: $[-4, -2]$

⑨ Domain

Words: All real numbers greater than -5.

Set Notation: $\{x \mid x > -5\}$

Interval Notation: $(-5, \infty)$



Range

Words: All real numbers greater than or equal to -3.

Set Notation: $\{y \mid y \geq -3\}$

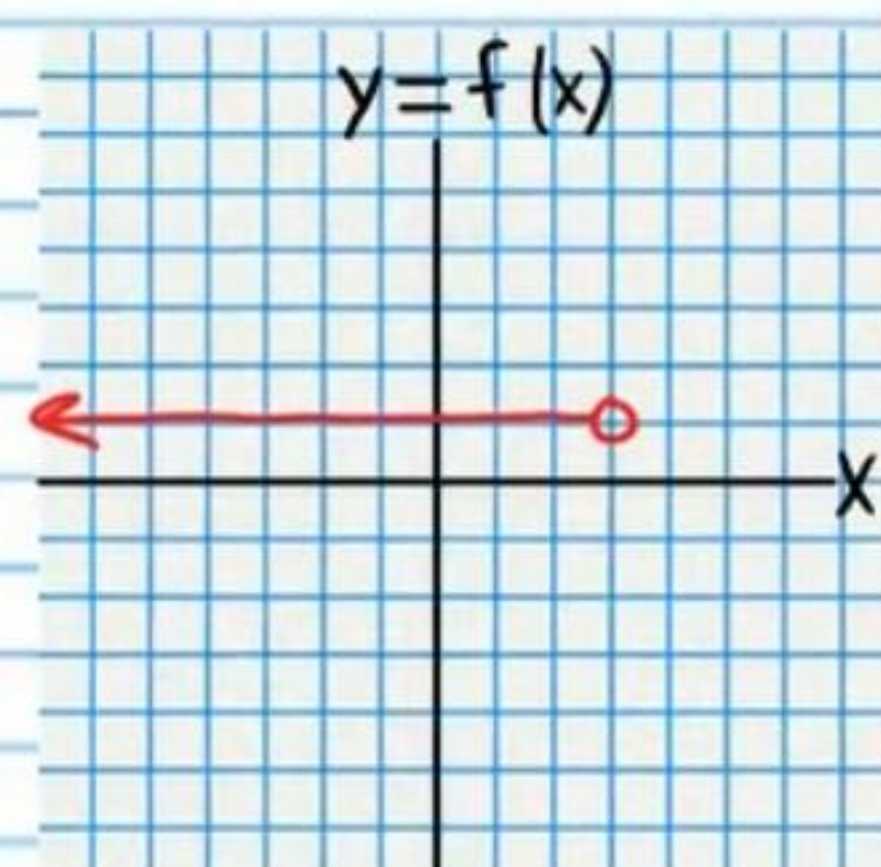
Interval Notation: $[-3, \infty)$

⑩ Domain

Words: All real numbers less than 3.

Set Notation: $\{x \mid x < 3\}$

Interval Notation: $(-\infty, 3)$



Range

Words: The only number in the range is 1.

Set Notation: $\{y \mid y = 1\}$

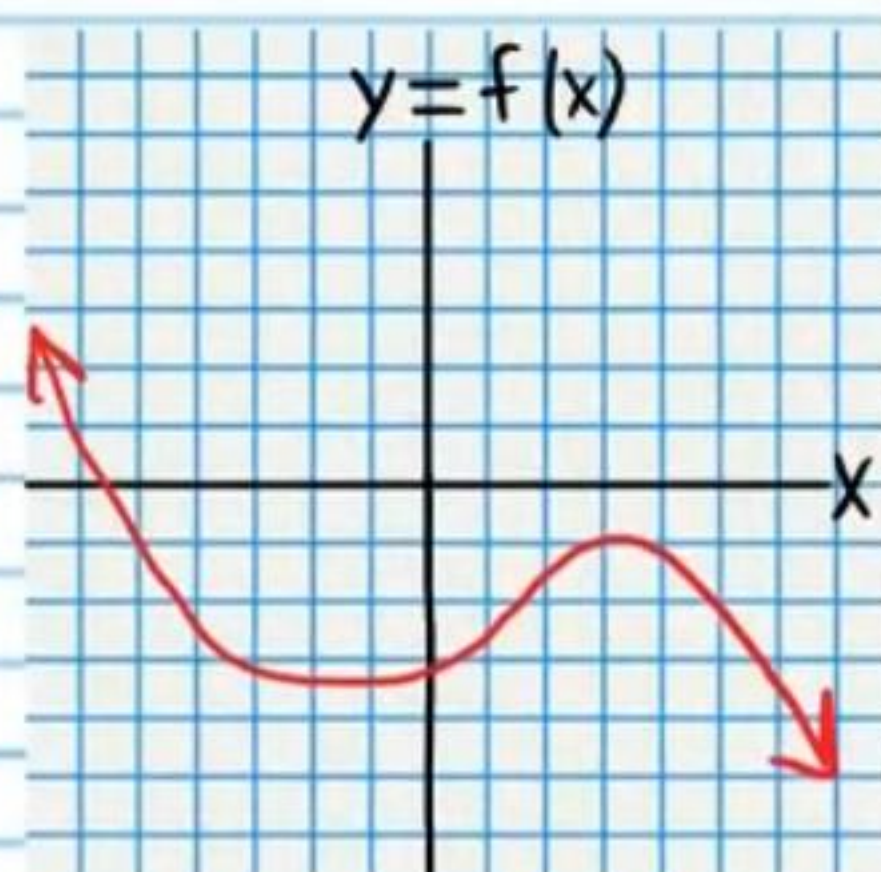
Interval Notation: $[1, 1]$

⑪ Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$



Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

⑫ Domain

Words: All real numbers.

Set Notation: All real numbers.

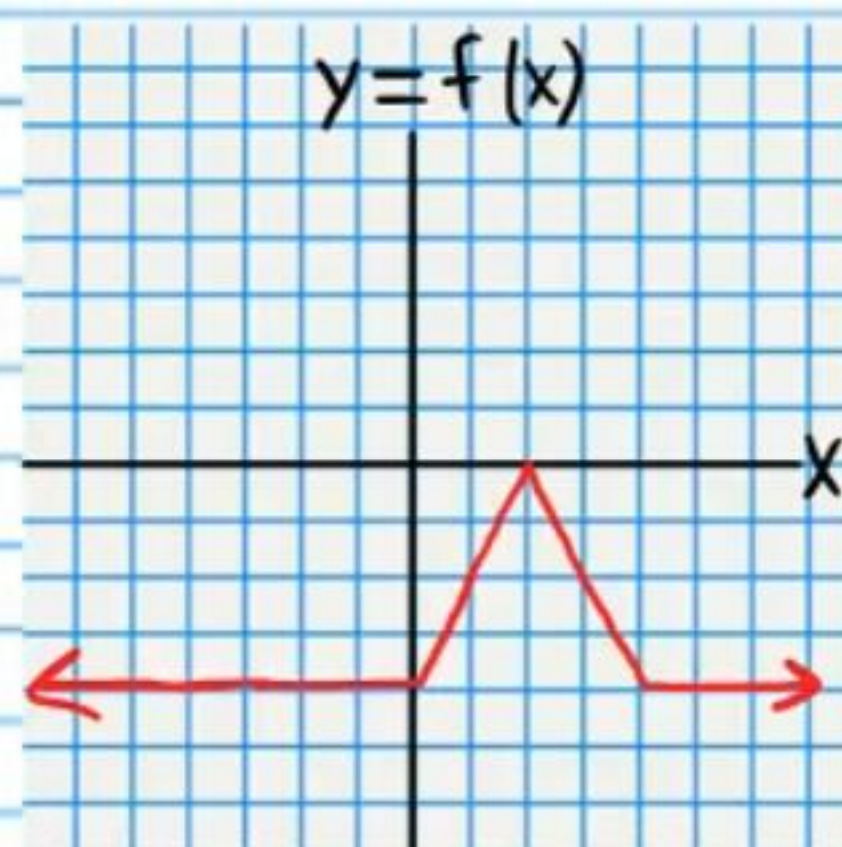
Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers between -4 and zero, inclusive.

Set Notation: $\{y \mid -4 \leq y \leq 0\}$

Interval Notation: $[-4, 0]$



⑬ Domain

Words: All real numbers less than -3 or greater than zero.

Set Notation: $\{x \mid x < -3 \text{ or } x > 0\}$

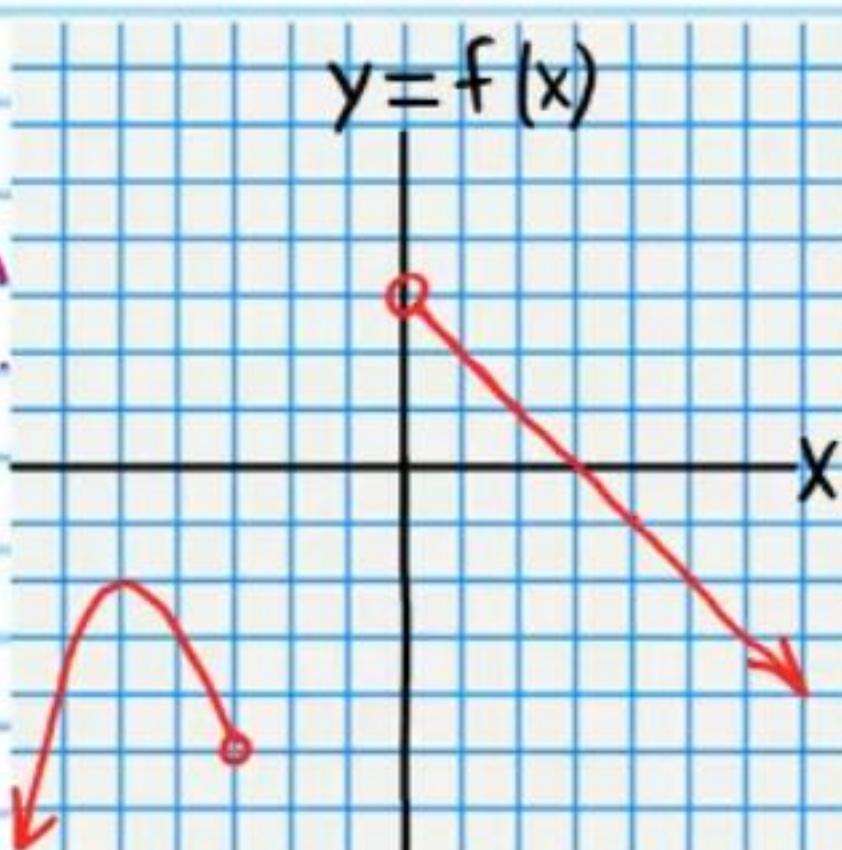
Interval Notation: $(-\infty, -3) \cup (0, \infty)$

Range

Words: All real numbers less than 3.

Set Notation: $\{y \mid y < 3\}$

Interval Notation: $(-\infty, 3)$



⑭ Domain

Words: All real numbers less than 1 or greater than 3.

Set Notation: $\{x \mid x < 1 \text{ or } x > 3\}$

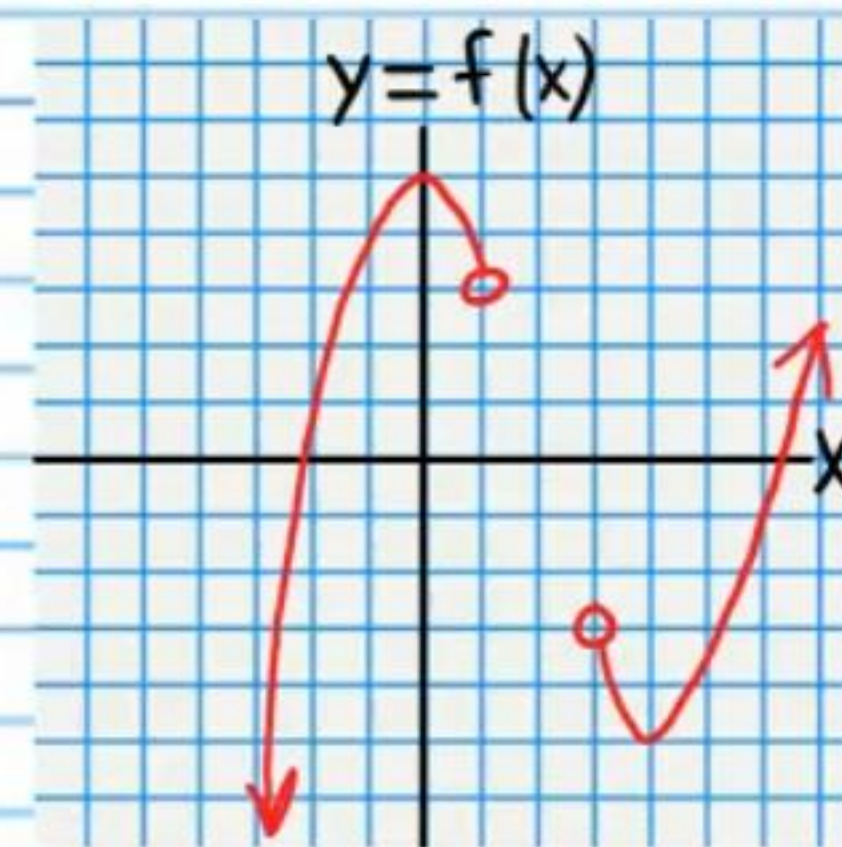
Interval Notation: $(-\infty, 1) \cup (3, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$



⑮ Domain

Words: All real numbers less than -2 or greater than 1.

Set Notation: $\{x \mid x < -2 \text{ or } x > 1\}$

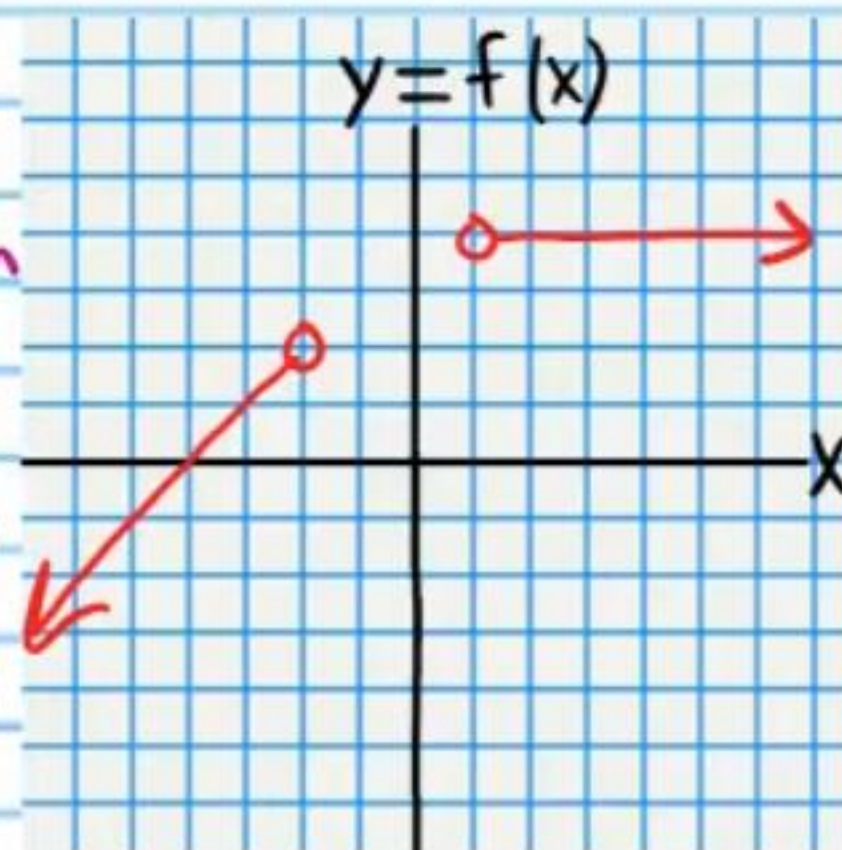
Interval Notation: $(-\infty, -2) \cup (1, \infty)$

Range

Words: 4 or all real numbers less than 2.

Set Notation: $\{y \mid y < 2 \text{ or } y = 4\}$

Interval Notation: $(-\infty, 2) \cup [4, 4]$



⑩ Domain

Words: All real numbers except -3.

Set Notation: $\{x | x \neq -3\}$

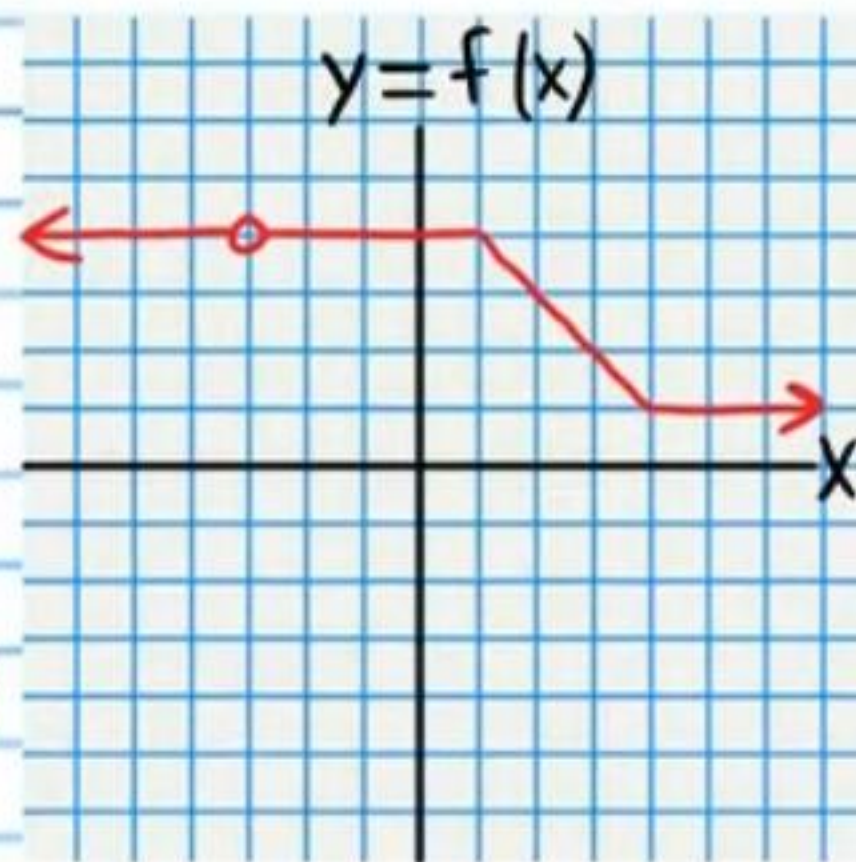
Interval Notation: $(-\infty, -3) \cup (-3, \infty)$

Range

Words: All real numbers between 1 and 4, inclusive.

Set Notation: $\{y | 1 \leq y \leq 4\}$

Interval Notation: $[1, 4]$



⑪ Domain

Words: All real numbers except 1 and 5.

Set Notation: $\{x | x \neq 1 \text{ and } x \neq 5\}$

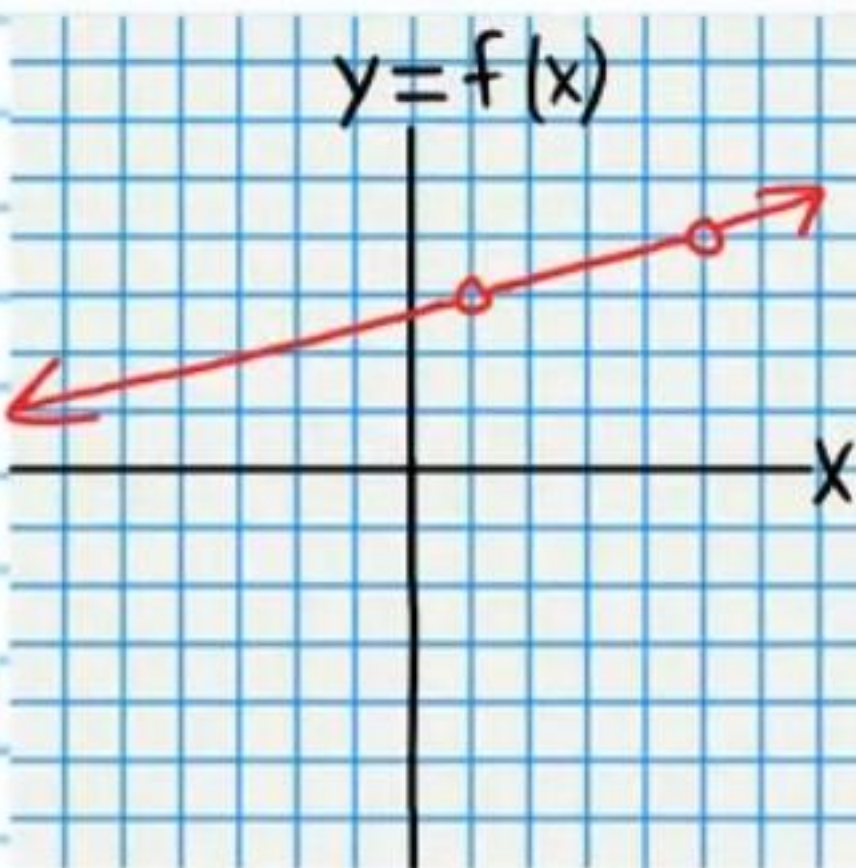
Interval Notation: $(-\infty, 1) \cup (1, 5) \cup (5, \infty)$

Range

Words: All real numbers except 3 and 4.

Set Notation: $\{y | y \neq 3 \text{ and } y \neq 4\}$

Interval Notation: $(-\infty, 3) \cup (3, 4) \cup (4, \infty)$



⑫ Domain

Words: All real numbers except -1.

Set Notation: $\{x | x \neq -1\}$

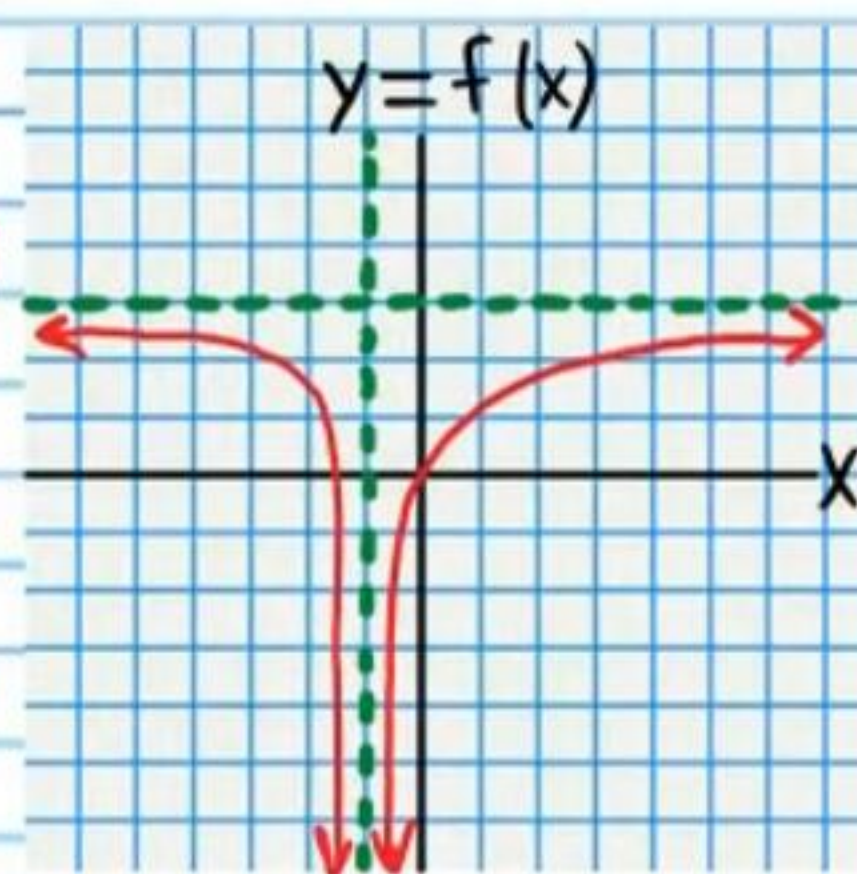
Interval Notation: $(-\infty, -1) \cup (-1, \infty)$

Range

Words: All real numbers less than 3.

Set Notation: $\{y | y < 3\}$

Interval Notation: $(-\infty, 3)$



⑬ Domain

Words: All real numbers except -5.

Set Notation: $\{x | x \neq -5\}$

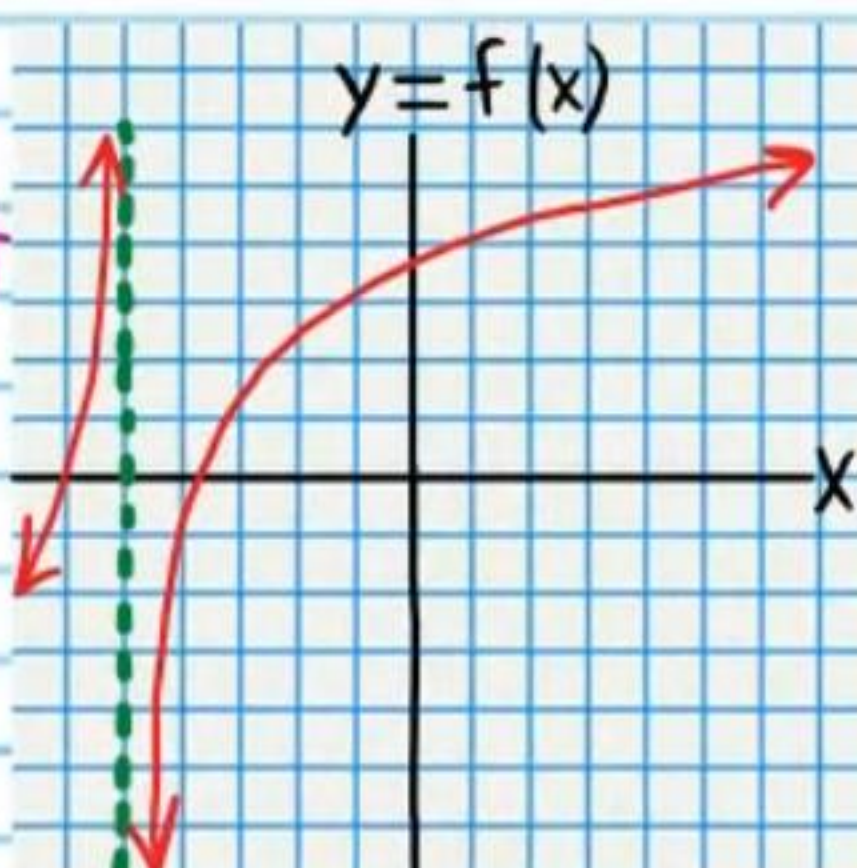
Interval Notation: $(-\infty, -5) \cup (-5, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$



(20) Domain

Words: All real numbers except 0, 2, and 5.

Set Notation: $\{x \mid x \neq 0, 2, \text{ and } 5\}$

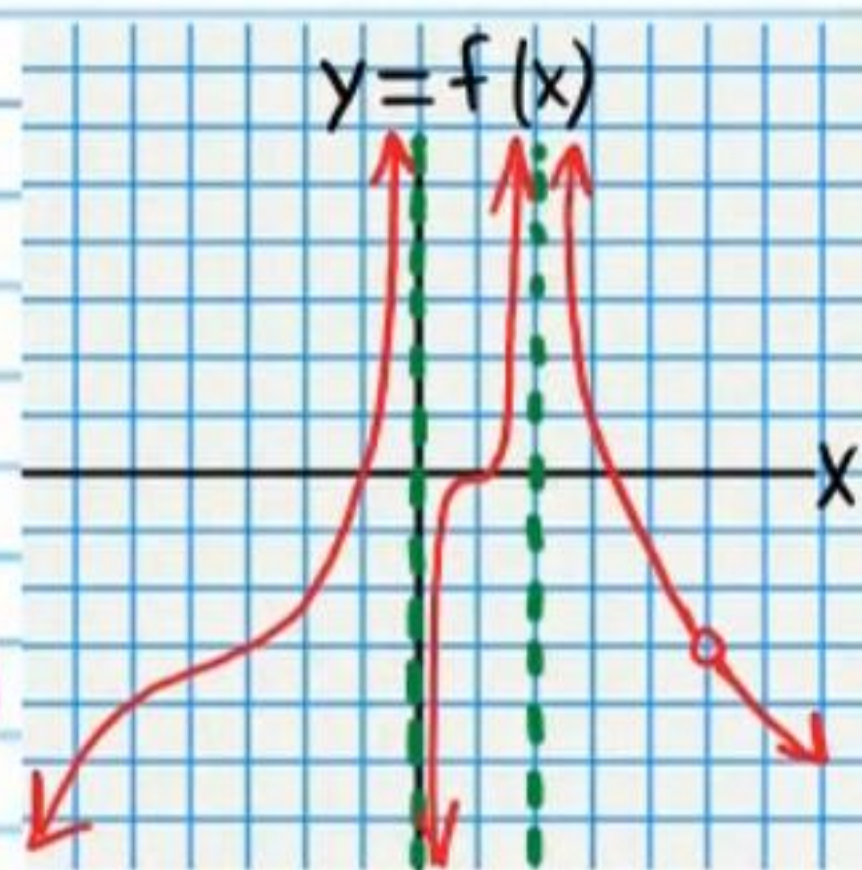
Interval Notation: $(-\infty, 0) \cup (0, 2) \cup (2, 5) \cup (5, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$



(21) Domain

Words: All real numbers.

Set Notation: All real numbers.

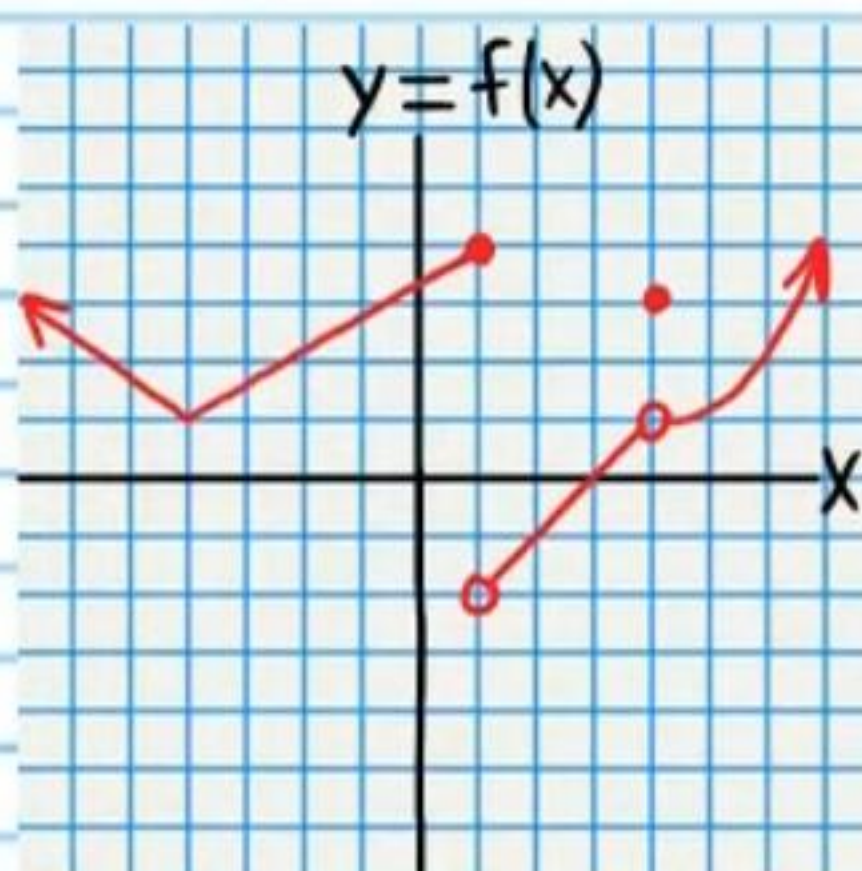
Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers greater than -2.

Set Notation: $\{y \mid y > -2\}$

Interval Notation: $(-2, \infty)$



(22) Domain

Words: All real numbers except 1.

Set Notation: $\{x \mid x \neq 1\}$

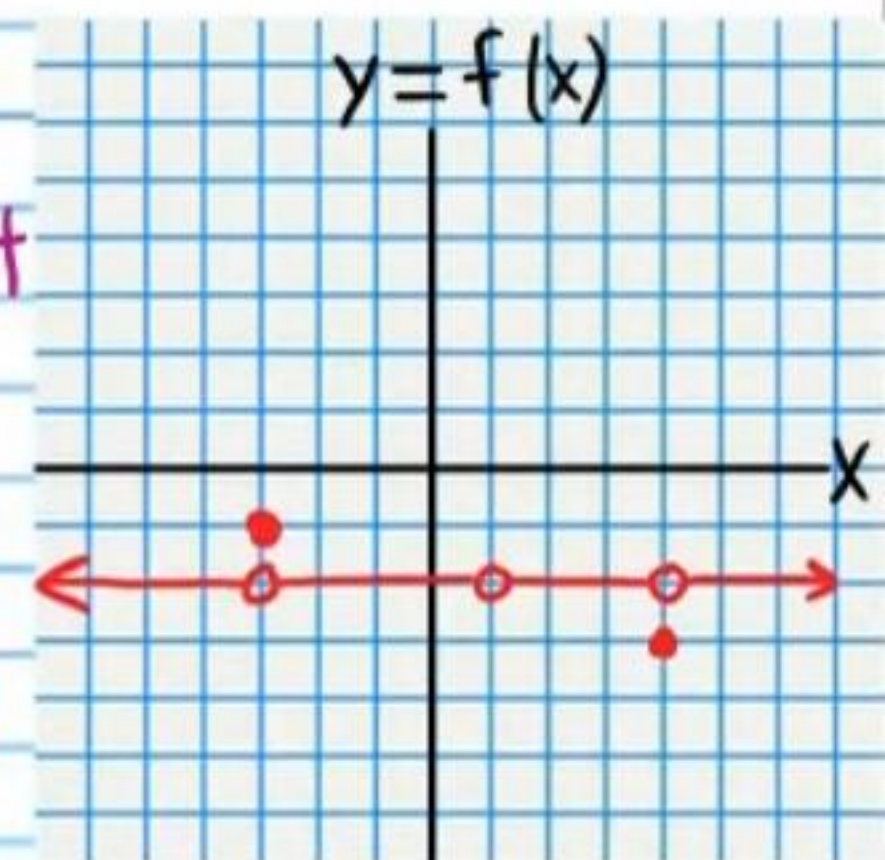
Interval Notation: $(-\infty, 1) \cup (1, \infty)$

Range

Words: The only numbers in the range are -3, -2, and -1.

Set Notation: $\{y \mid y = -3, -2, \text{ or } -1\}$

Interval Notation: $[-3, -3] \cup [-2, -2] \cup [-1, -1]$



(23) Domain

Words: The only numbers in the domain are -6, 1, and 6.

Set Notation: $\{x \mid x = -6, 1, \text{ or } 6\}$

Interval Notation: $[-6, -6] \cup [1, 1] \cup [6, 6]$

Range

Words: The only numbers in the range are -4 and 1.

Set Notation: $\{y \mid y = -4 \text{ or } y = 1\}$

Interval Notation: $[-4, -4] \cup [1, 1]$

